FREE REVISION HANDBOOK

Every GRE Quant formula, rule and trap. One document.

Nine sections, every formula worth memorizing, the traps that actually cost marks, and an appendix of the values you should never have to work out. Free to read, free to print, and built from the same notes my students use all year.

How to use this handbook

This is not a textbook. It is the last thing you read before a mock, and the last thing you read before the real exam. Nothing here teaches a topic from scratch — every line assumes you have met the idea before and need it back in working memory fast.

Read it in one of three ways, depending on how much time you have.

YOU HAVE	READ	WHY
90 minutes	The whole handbook, front to back, once	The full revision pass. Do this the day before a mock.
25 minutes	Every **Formulas and rules** box, skipping Traps and Speed	The formula-only pass. Do this the morning of a mock.
10 minutes	Section A, plus Section 9 (the 25 costly traps), plus the Appendix	The night-before pass. No new learning, only recall.

The one rule for using it well. When a line surprises you, stop and open the full concept note for that topic. A formula you cannot immediately explain is a formula you will misapply under time pressure. Surprise is the signal that you need the note, not the handbook.

Notation used throughout. Mathematics is written in plain text so this reads correctly on paper, on a phone and in print: `sqrt(x)` is the square root of x, `x^2` is x squared, `pi` is about 3.14, `|x|` is the absolute value of x, and `a/b` is a divided by b. Multiplication is written with `*` where it could be ambiguous, and by juxtaposition otherwise.

Section A — The test itself

Know this cold. These are marks you lose to the format, not to the mathematics.

A.1 Structure and timing

SECTION	QUESTIONS	TIME	AVERAGE PER QUESTION
Quant 1	12	21 minutes	about 1 minute 45 seconds
Quant 2	15	26 minutes	about 1 minute 44 seconds

- **No penalty for a wrong answer.** Never leave a blank. A guess is strictly better than nothing.
- **Adaptive between sections, not within.** How you do on the first Quant section sets the difficulty of the second. Within a section, difficulty tells you nothing about how you are doing.
- **You may move freely inside a section.** Flag a question, finish the section, come back.
- **An on-screen calculator is provided.** It is a four-function calculator with a square root. It is slower than your head for small arithmetic and it does not respect order of operations the way you expect. Use it for long division, awkward decimals and square roots, and for nothing else.

A.2 The real time budget

The average is misleading, because the hard questions need more than the average and the easy ones need far less. Work to these targets instead.

QUESTION TYPE	TARGET
Quantitative comparison	60 to 75 seconds
Straightforward multiple choice	60 to 90 seconds
Data interpretation, per question	60 to 90 seconds, plus 40 seconds to read the set first
Hard multi-step problem	up to 2 minutes, then move on

Banking 30 seconds on each of six easy questions buys three extra minutes for the hard ones. That trade is the whole game.

The 30-second rule. If you cannot see a route into a problem within 30 seconds, flag it and move on. Every question is worth the same mark. There is no prize for answering them in order.

Two passes, always. First pass: everything you can do at speed. Second pass: the flagged ones.

A.3 The four question formats

FORMAT	HOW YOU RECOGNIZE IT	THE RULE THAT MATTERS
Multiple choice, one answer	5 options, **oval** buttons	Exactly one is correct. Back-solving is available.
Multiple choice, one or more	3 to 8 options, **square** boxes	Select every correct one. **No partial credit.**
Numeric entry	A box, or two boxes for a fraction	No options, so no elimination and no safety net.
Quantitative comparison	The standard four choices	Compare; do not compute.

Oval buttons mean pick one. Square boxes mean pick all that apply. Check the shape before you answer. It takes one second and it prevents a full-mark loss.

On multiple-answer questions: there is no partial credit, so two right out of three scores zero. The number of correct options is usually not stated. Walk down the list once, treating each option as its own true-or-false question, and do not stop when you have found two.

On numeric entry: your work has to be right, because nothing on screen will catch you. Budget an extra 15 seconds to sanity-check. Enter a fraction in the two-box form when it is offered, and it need not be reduced. Follow any rounding instruction exactly. Do not type units, commas or symbols. Use the minus sign in the box for a negative answer.

A.4 Quantitative comparison — the method

Roughly a third to a half of every Quant section, so expect somewhere around five or six in the first section and six or seven in the second. The four choices are always the same, in this order:

```
Quantity A is greater.
Quantity B is greater.
The two quantities are equal.
The relationship cannot be determined from the information given.
```

The decision rule for the fourth choice. Choose it if and only if you can produce **two cases that give different answers**. One case with Quantity A greater and one case with the two equal is enough.

When the fourth choice is impossible. If both quantities are pure numbers with no variable in sight, the relationship is fixed, even if you cannot yet see it. The answer can never be the fourth choice.

Operations you may perform on both quantities. They behave exactly like inequalities.

OPERATION	SAFE?
Add or subtract the same value	Yes
Multiply or divide by the same **positive** value	Yes
Multiply or divide by a **negative** value	No — it reverses the relationship
Multiply or divide by an expression of unknown sign	No
Square both quantities	Only when both are known to be non-negative

The FROZEN test values. When a variable is unrestricted, test these before concluding anything:

```
F  Fractions        (try 0.5)
R  Repeats          (make the variables equal)
O  One
Z  Zero
E  Extremes         (very large, very small)
N  Negatives
```

If two of them disagree, the answer is the fourth choice. If they all agree, the answer is probably one of the first three — but test at least three values before you commit.

Speed on QC. Strip away whatever is common to both quantities and compare what is left; the remainder is usually trivial. Estimate rather than compute whenever the gap is wide. If the two quantities look designed to be equal, suspect "equal" and then test one edge case to confirm.

A.5 Test-day rules that are not mathematics

- **Read the last line again before you answer.** Solving for x and then answering with x when the question asked for 2x is the single most common avoidable error on the test.
- **Underline what is asked** before you start solving.
- **Write down every intermediate value.** Re-deriving a number you already had is pure waste, and mental algebra under time pressure produces sign errors.
- **Sanity-check the shape of the answer.** A negative length, a probability above 1, a percentage over 100, an average outside the range of the data: each of these means you have made an error.
- **Do not assume a figure is drawn to scale** unless the question says so. In quantitative comparison especially, many are deliberately not.
- **A hard question means nothing.** Difficulty within a section is not feedback. Move on.

Section 1 — Arithmetic

1.1 Percentages

Every percent question is really the question *percent of what*. Errors here are almost never arithmetic; they are a wrong choice of base.

Formulas and rules

NEED	METHOD
x percent of y	`(x/100) * y`
Find the part	multiply: 15 percent of 80 is `0.15 * 80`
Find the whole	divide: 12 is 15 percent of what, so `12 / 0.15`
Find the percent	`(part / whole) * 100`
Percent change	`(New - Old) / Old * 100` — always divide by the **original**
Increase by x percent	multiply by `1 + x/100`
Decrease by x percent	multiply by `1 - x/100`
Successive changes	multiply the multipliers: `1.40 * 0.80 * 0.90`
Reverse a change	divide by the multiplier, do not subtract the percent

Turn every percent into a multiplier, then multiply. One line instead of three calculations.

The six traps

TRAP	WRONG	RIGHT
New value used as the base	50 to 60 is `10/60` = 16.7 percent	50 to 60 is `10/50` = 20 percent
Adding successive percents	up 30 then down 30 is 0	`1.30 * 0.70` = 0.91, a 9 percent loss
"More than" read as "of"	A is 40 percent more than B, so `A = 0.40B`	`A = 1.40B`
Percent against percentage points	5 percent to 7 percent is "a 2 percent rise"	2 percentage points, but a 40 percent rise
Percent of a percent	20 percent of 30 percent is 10 percent	`0.20 * 0.30` = 6 percent
Equal fall and rise assumed to cancel	down 20 then up 20 returns to the start	`0.80 * 1.20` = 0.96, still 4 percent down

Order does not matter in a multiplier chain: `0.8 * 1.2` and `1.2 * 0.8` both give 0.96. A discount then a tax equals a tax then a discount, so never recompute it.

Speed

- **Pick 100.** When the problem is all percents and no amounts, set the start value to 100.
- **Split the percent.** 15 percent of 240 is 10 percent plus 5 percent, so `24 + 12 = 36`.
- **Swap the terms.** x percent of y equals y percent of x. So 8 percent of 25 is 25 percent of 8, or 2.
- **One percent, then scale.** 17 percent of 350: one percent is 3.5, so `3.5 * 17 = 59.5`.
- **Percent as a fraction.** 12.5 percent is 1/8, so dividing by 8 beats multiplying by 0.125.
- **Predict the direction first.** Decide whether the answer is bigger or smaller than the start. Half of all trap answers fail that check instantly.

1.2 Fractions and decimals

Formulas and rules

OPERATION	RULE
`a/b + c/d`	`(ad + bc) / bd`
`a/b - c/d`	`(ad - bc) / bd`
`a/b * c/d`	`ac / bd`
`a/b` divided by `c/d`	`a/b * d/c` — flip the **second** one
`(a/b)^n`	`a^n / b^n`
Compare `a/b` with `c/d`	cross-multiply: `ad` against `bc`, valid when b and d are positive
Mixed to improper	`a b/c = (ac + b)/c`

Size intuition

- Same numerator, bigger denominator gives a **smaller** fraction: `3/7 < 3/5`.
- Same denominator, bigger numerator gives a **bigger** fraction: `5/7 > 3/7`.
- Adding the same positive k to the top and bottom of a **proper** fraction moves it **toward 1**, so it gets bigger: `2/3 < 3/4 < 4/5`.
- For an **improper** fraction, adding k to both moves it **down** toward 1: `5/3 > 6/4 > 7/5`.

Terminating or recurring. In lowest terms, a fraction terminates if and only if the denominator's prime factorization contains **only 2s and 5s**. `7/40 = 7/(2^3 * 5)` terminates at 0.175; `5/12 = 5/(2^2 * 3)` recurs as 0.41666...

Decimals. Multiplying: multiply as integers, then the answer carries as many decimal places as the two factors combined, so `0.03 * 0.4` gives `12` with three places, or 0.012. Dividing: shift both points right until the divisor is a whole number. And `0.999...` equals 1 exactly.

Traps

- **Adding across.** `1/2 + 1/3` is not `2/5`.
- **Not reducing before judging a denominator.** `9/24` looks like it has a 3 on the bottom; reduced to `3/8` it does not, so it terminates.
- **Flipping the wrong fraction** when dividing. Flip the divisor.
- **Squaring a fraction between 0 and 1 makes it smaller.** `(1/2)^2 = 1/4`. The square root of a number between 0 and 1 is **larger** than the number.
- **Negative fractions reverse intuition.** `-1/2 > -3/4`, because it is closer to zero.
- **Decimal place miscount.** `0.2 * 0.3` is 0.06, not 0.6.

Speed

- **Stay in fractions** until the final step, especially on numeric entry, which accepts fractions.
- **Cancel before multiplying.** `(14/25) * (75/28)`: cancel 14 against 28 and 75 against 25 to get `3/2`.
- **Cross-multiply to compare.** It beats converting to decimals every time.
- **Benchmark against 1/2 and 1** for fast triage.
- **Recurring decimals worth knowing:** `1/3 = 0.333`, `1/6 = 0.1666`, `1/7 = 0.142857` repeating, `1/9 = 0.111`, `1/11 = 0.0909`.

1.3 Ratios and proportions

A ratio is not an amount. Write `ak` and `bk`, then use the extra fact in the stem to find k.

Formulas and rules

CONCEPT	RULE
Part to whole from `a : b`	first part is `a/(a+b)` of the total
Introducing the multiplier	`a : b` means the amounts are `ak` and `bk`
Sum of parts	`a + b + c` divides the total; if it does not divide cleanly, re-read the stem
Direct proportion	`y = kx`; one rises, the other rises
Inverse proportion	`xy = k`; one rises, the other falls
Chaining `A:B` with `B:C`	scale so the shared term matches
Similar figures, side ratio k	lengths `k`, areas `k^2`, volumes `k^3`
Comparing `a : b` with `c : d`	cross-multiply, as with fractions

Chaining worked. `A:B = 2:3` and `B:C = 4:5`. B is 3 then 4, so scale to 12: `A:B = 8:12` and `B:C = 12:15`, giving `A:B:C = 8:12:15`.

The six traps

TRAP	WRONG	RIGHT
Part-to-part read as part-to-whole	boys : girls `= 2:3`, so boys are `2/3` of the class	boys are `2/5`
Changing both parts when only one moved	`(2k+10)/(3k+10) = 4/3`	`(2k+10)/(3k) = 4/3`
Chaining without matching the shared term	`A:B:C = 2:3:5`	`8:12:15`
Reading the ratio as the amounts	`3:5` means 3 boys and 5 girls	it means `3k` and `5k`
Direct used where inverse belongs	12 workers take `15 * 12/8` days	12 workers take `15 * 8/12` = 10 days
Forgetting to square or cube	sides `2:3`, so areas `2:3`	areas `4:9`, volumes `8:27`

Speed

- **Always name k**, even on easy problems. It turns a ratio problem into an equation problem.
- **Reduce the ratio before introducing k.** `6:10:16` and `3:5:8` give the same shares with half the arithmetic.
- **Predict the direction.** "More workers, fewer days" kills the direct-against-inverse trap with no algebra at all.
- **Track the quantity that does not change** in any mixture or change problem — usually the pure solute, or the group nobody joined or left.
- **Re-read the final sentence.** Ratio problems routinely ask for the amount *added*, or for B when you solved for k.

1.4 Exponents and roots

Formulas and rules

RULE	STATEMENT
Product	`a^m * a^n = a^(m+n)`
Quotient	`a^m / a^n = a^(m-n)`
Power of a power	`(a^m)^n = a^(mn)`
Power of a product	`(ab)^n = a^n b^n`
Power of a quotient	`(a/b)^n = a^n / b^n`
Zero exponent	`a^0 = 1` for a not equal to 0
Negative exponent	`a^(-n) = 1/a^n`, for a not equal to 0
Fractional exponent	`a^(m/n)` is the n-th root of `a^m`
Root as a power	`sqrt(a) = a^(1/2)`, cube root of a is `a^(1/3)`

What the rules do not say

- `a^m + a^n` does **not** become `a^(m+n)`. Addition has no exponent rule. Factor instead: `2^5 + 2^6 = 2^5(1 + 2) = 3 * 2^5`.
- `(a + b)^2` is **not** `a^2 + b^2`. It is `a^2 + 2ab + b^2`.
- `sqrt(a^2 + b^2)` is **not** `a + b`.

Signs. `(-2)^4 = 16` (even power, positive). `(-2)^3 = -8` (odd power, negative). But `-2^4 = -16`, because the minus sits outside and only the 2 is raised.

Roots. The symbol `sqrt(x)` denotes the **non-negative** root, so `sqrt(25) = 5` and never -5. But the *equation* `x^2 = 25` has two solutions, `x = 5` and `x = -5`. This distinction is tested constantly. Also `sqrt(x^2) = |x|`, not x.

Simplifying. Pull out perfect squares: `sqrt(72) = sqrt(36 * 2) = 6 sqrt(2)`. **Rationalizing.** `1/sqrt(3) = sqrt(3)/3`. For `1/(2 + sqrt(3))`, multiply above and below by the conjugate `2 - sqrt(3)` to get `2 - sqrt(3)`.

Traps

- **`(-2)^2` against `-2^2`.** With brackets, 4. Without, -4.
- **`x^2 = 16` gives `x = 4` only.** No: `x = 4` or `x = -4`. Missing the negative root is the number-one exponent error on quantitative comparison.
- **Adding exponents across different bases.** `2^3 * 3^3` is `6^3`, not `6^6`.
- **Fractional bases shrink.** `(1/2)^5` is *smaller* than `(1/2)^2`.
- **`a^(1/2)` read as half of a.** It is the square root.
- **A negative exponent read as a negative number.** `2^(-3) = 1/8`.

Speed

- **Prime-factorize everything.** Any exponent equation becomes trivial once every base is a prime power.
- **Factor, do not expand,** whenever you see a sum of powers with the same base.
- **To compare `a^m` with `b^n`, force a common base or a common exponent** — whichever is easier.
- **Bracket roots.** `sqrt(50)` lies between 7 and 8 and close to 7, so about 7.07.
- **Units-digit cycles** repeat with period at most 4: 2 gives 2,4,8,6; 3 gives 3,9,7,1; 7 gives 7,9,3,1; 8 gives 8,4,2,6. The digits 0, 1, 5 and 6 never change.

1.5 Rates, work and unit conversion

Formulas and rules

CONCEPT	FORMULA
Distance	`Speed * Time`
Work done	`Rate * Time`
Working together	add the rates: `R1 + R2`
Fill against drain	subtract: `R1 - R2`
Average speed	`Total distance / Total time` — always
Equal **distances** at speeds a and b	`2ab/(a + b)`
Equal **times** at speeds a and b	`(a + b)/2`
Same direction	relative speed is the **difference**
Opposite directions	relative speed is the **sum**
Unit conversion	multiply by a fraction equal to 1, `desired/given`

Average speed is never the average of the speeds unless the *times* are equal.

Conversions. 1 hour is 3600 seconds and 1 kilometer is 1000 meters. To convert kilometers per hour to meters per second, multiply by `5/18`; to go back, multiply by `18/5`. One mile is about 1.6 kilometers and 1 kilogram is about 2.2 pounds, though the GRE normally supplies any non-metric factor.

Traps

- **Averaging the speeds** on an equal-distance round trip. That is the planted wrong answer.
- **Not converting units first.** Speed in kilometers per hour with time in minutes needs a conversion before you multiply.
- **Adding times instead of rates.** Two people taking 6 hours and 12 hours do not together take 18 hours, nor 9 by averaging. They take 4. Add their rates.
- **Forgetting that a leak subtracts.** Rates add only when both agents push the same way.
- **"50 percent faster" is not "half the time".** Rate times 1.5 means time times 2/3.
- **Rounding mid-problem.** `20/3` hours is exact; 6.67 invites error.

Speed

- **Use the least common multiple as the total work.** For workers taking 6 and 12 hours, set the job at 12 units: A does 2 units an hour, B does 1, together 3, so `12/3 = 4` hours. No fractions at all.
- **Chain the units** as a product of fractions and cancel diagonally. If the leftover units are right, the arithmetic is right.
- **Memorize `* 5/18`** for kilometers per hour into meters per second.
- **For "when do they meet", add the speeds** and divide the gap by that sum.
- **Predict whether the answer sits nearer the faster or the slower value.** This catches trap answers instantly.

1.6 Remainders and cyclicity

Formulas and rules

CONCEPT	RULE
Division statement	`N = dq + r` with `0 <= r < d`
Remainder range	always less than the divisor; 0 is allowed and means divisible
Sum	the remainder of `a + b` is the remainder of (remainder of a plus remainder of b)
Product	the remainder of `a * b` is the remainder of (remainder of a times remainder of b)
Power	reduce the base to its remainder first, then use the cycle
Last digit	the remainder on division by 10
Reading the stem	"leaves remainder 3 when divided by 7" means `N = 7k + 3`

Units-digit cycles for powers

UNITS DIGIT OF THE BASE	CYCLE	LENGTH
0, 1, 5, 6	itself	1
4	4, 6	2
9	9, 1	2
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4

For a cycle of length 4, divide the exponent by 4 and read the remainder. A remainder of 0 means use the **fourth** entry, not the first.

Traps

- **Treating remainder 0 as invalid.** Zero is a perfectly good remainder.
- **Leaving a remainder at or above the divisor.** A remainder of 9 on division by 7 reduces to 2.
- **Adding remainders without re-reducing.** 4 plus 5 against a divisor of 7 gives 9, which reduces to 2.
- **Negative remainders.** The GRE uses non-negative remainders: -3 divided by 5 leaves 2.
- **Off-by-one on cycles.** For `2^12`, the exponent leaves remainder 0 on division by 4, so take the fourth entry, 6.
- **Assuming a unique answer.** "n leaves remainder 3 on division by 7" describes infinitely many n, so on quantitative comparison test at least two of them.

Speed

- **Write `N = dq + r` immediately.** Nearly every remainder word problem becomes algebra once you do.
- **Pick the smallest valid number,** then one more to confirm.
- **Reduce before you multiply.** For `97 * 98` against a divisor of 5, use `2 * 3 = 6`, which leaves 1.
- **Same remainder for two divisors** points to the least common multiple plus r.
- **Last two digits** work the same way, against a divisor of 100.

1.7 Absolute value

Read `|x|` as a distance from zero, and `|a - b|` as the distance between a and b.

Formulas and rules

STATEMENT	MEANING
`|x| = x` when `x >= 0`, `-x` when `x < 0`	and `-x` is positive in that second case
`|x| >= 0` always	
`|-x| = |x|`	
`|ab| = |a| * |b|` and `|a/b| = |a|/|b|`	the bars distribute over times and divide
`|a + b| <= |a| + |b|`	they do **not** distribute over plus
`|x| = k` with `k > 0`	`x = k` or `x = -k`, two solutions
`|x| = 0`	`x = 0`, one solution
`|x| = k` with `k < 0`	no solution
`|x| < k` with `k > 0`	`-k < x < k`, one interval
`|x| > k` with `k > 0`	`x < -k` or `x > k`, two rays
`sqrt(x^2)`	equals `|x|`, not x

The general pattern. `|x - c| < r` says x is within r of c, which is the interval from `c - r` to `c + r`. So `|x - 3| < 5` gives `-2 < x < 8`.

Traps

- **Taking only the positive case.** `|x| = 5` has two answers.
- **Treating `|x| > k` like `|x| < k`.** Less-than is one interval; greater-than is two rays.
- **Distributing the bars over a sum.** With `a = 3` and `b = -3`, `|a + b| = 0` but `|a| + |b| = 6`.
- **Assuming `sqrt(x^2) = x`.** It is `|x|`.
- **Skipping the extraneous-solution check** when the variable also appears outside the bars, as in `|x - 2| = 3x`. Solve both cases, substitute back, discard whatever fails.
- **Forgetting to flip an inequality** when you multiply or divide by a negative.

Speed

- **Read it as distance.** `|x - 7| = 2` means x is 2 away from 7, so `x = 5` or `x = 9`. No algebra.
- **Sketch the number line.** Mark the center and the radius; the answer is inside or outside.
- **On quantitative comparison, test a negative** whenever a variable inside bars is unrestricted. That is where the trap lives.
- **`|a| = |b|` means `a = b` or `a = -b`.** Two cases, always.
- **The least value of `|x - a| + |x - b|` is `|a - b|`**, reached for any x between a and b.

1.8 Powers of 10 and scientific notation

Formulas and rules

CONCEPT	RULE
Standard form	`a * 10^n` with `1 <= a < 10`
Multiplying	`(a * 10^m)(b * 10^n) = ab * 10^(m+n)`, then renormalize
Dividing	`(a * 10^m)/(b * 10^n) = (a/b) * 10^(m-n)`
Adding	match the exponents first, then add the coefficients
Point moves right	the exponent drops by 1 per place
Point moves left	the exponent rises by 1 per place
`10^0`	1
`10^(-n)`	`1/10^n`, a small positive number

Place value. `10^3` is a thousand, `10^6` a million, `10^9` a billion, `10^(-2)` a hundredth and `10^(-3)` a thousandth.

Comparing. Compare exponents first and only break a tie on the coefficient. So `3.1 * 10^8 > 9.9 * 10^7`.

Traps

- **`10^0 = 0`.** It is 1.
- **A negative exponent read as a negative number.** `10^(-4) = 0.0001`, positive and small.
- **Adding coefficients without matching exponents.** `3 * 10^5 + 2 * 10^3` is neither `5 * 10^8` nor `5 * 10^5`.
- **Forgetting to renormalize.** Answer choices are always in proper standard form.
- **Miscounting decimal places.** `0.0034` is `3.4 * 10^(-3)`. Count from the point to the first non-zero digit.
- **Assuming more digits means bigger.** `1.9 * 10^7` is far smaller than `2.0 * 10^9`.

Speed

- **Split every number into coefficient and power,** then handle the two parts separately.
- **For "approximately" questions, round the coefficient to one digit and just track the exponent.**
- **For "how many times greater", divide** — the answer falls out as a power of 10.
- **Check the sign of the exponent:** above 1 is positive, between 0 and 1 is negative.

1.9 Number line and ordering

Formulas and rules

CONCEPT	RULE
Ordering	further right is greater
Midpoint of a and b	`(a + b)/2` — an average, not a difference
Distance between a and b	`|a - b|`
Negatives	closer to zero is greater, so `-1/2 > -1`
Squaring with `0 < x < 1`	`x^2 < x`, it moves toward 0
Squaring with `x > 1`	`x^2 > x`
Square root with `0 < x < 1`	`sqrt(x) > x`
Square root with `x > 1`	`sqrt(x) < x`
Reciprocal with `x > 1`	`1/x` lands between 0 and 1
Reciprocal of a negative	still negative

The four regions to test. Whenever a variable is unrestricted, a complete check needs a value from each of `x < -1`, `-1 < x < 0`, `0 < x < 1` and `x > 1`, plus the boundaries 0, 1 and -1. Run it alongside the FROZEN list in Section A.4 — the two overlap, and between them they are what makes quantitative comparison safe.

Traps

- **`-3 > -2`.** No: more negative is smaller.
- **Assuming squaring enlarges.** Only outside the range -1 to 1.
- **`|a| > |b|` implies `a > b`.** False: take `a = -5` and `b = 3`.
- **Comparing fractions on the numerator alone.** `3/4` against `5/8` needs a common basis.
- **Forgetting 0 and 1 as test cases.** They break most "which is greater" patterns.
- **Assuming a variable is positive.** Unless the stem says so, it is not.

Speed

- **Convert mixed lists to decimals.** Three places is always enough on the GRE.
- **Benchmark against 1/2 and 1** before comparing precisely.
- **Draw it.** A ten-second number line prevents nearly every sign error.

1.10 Estimation and approximation

When to estimate

SITUATION	APPROACH
Answer choices far apart	round hard, then match
Answer choices close together	compute exactly
"Approximately" in the stem	estimation is intended — but check the spread
Percent of a large number	use 10 percent and scale
Square roots	bracket between perfect squares
Fractions	round to 1/2, 1/3 or 3/4

Controlling the direction of the error. This is the part most students miss. Rounding the numerator **up** and the denominator **down** makes your estimate too big twice over. Rounding both the **same** way lets the errors partly cancel. For `487/103`, true value 4.73, rounding to `500/100 = 5` overestimates twice, while `500/110 = 4.55` or `490/105 = 4.67` land much closer.

Root approximation. `sqrt(N)` sits between the two nearest perfect squares, roughly in proportion. For `sqrt(60)`: between 7 and 8, and 60 is 11 above 49 across a gap of 15, so about 7.7. The true value is 7.746.

Traps

- **Estimating when an exact value is wanted.** Check the answer spread first.
- **Compounding the error in a product.** In a quotient, rounding both parts the same way makes the two errors cancel; in a product it makes them pile up. Rounding 4.6 up to 5 and 2.4 up to 3 gives 15 against a true 11.04, a 36 percent miss.
- **Over-rounding percentages.** Treating 19.8 percent as 20 percent is fine; treating 17 percent as 20 percent usually is not.
- **Rounding before simplifying.** Cancel common factors first; the exact answer often becomes easy.
- **Estimating on quantitative comparison when the quantities are close.** If your estimates come out nearly equal, you must compute exactly — an estimate cannot separate "A is greater" from "equal".

Speed

- **Read the answer choices before calculating.** Their spread tells you the precision you need.
- **Ten percent, then scale.** 35 percent of 480 is `48 * 3.5 = 168`.
- **Double and halve.** `25 * 36 = 50 * 18 = 100 * 9 = 900`.
- **Percent as a fraction.** 12.5 percent is 1/8, 16.67 percent is 1/6, 37.5 percent is 3/8.
- **Check the order of magnitude last.** Even with shaky digits, getting `10^3` against `10^4` right usually identifies the answer.

Section 2 — Number Properties

2.1 Factors and multiples

Formulas and rules

CONCEPT	RULE
Factor	divides N exactly, leaving remainder 0
Multiple	`N * k` for an integer k
Smallest factor of N above 1	its smallest prime factor
Largest factor of N	N itself
Smallest positive multiple of N	N itself
Number of factors	if `N = p^a * q^b * r^c`, the count is `(a+1)(b+1)(c+1)`
Sum of factors	`(p^0 + ... + p^a)(q^0 + ... + q^b) * ...`
Factors come in pairs	d pairs with `N/d`
Perfect squares	have an **odd** number of factors
Count of multiples of k from a to b	`floor(b/k) - floor((a-1)/k)`

Why perfect squares have an odd count. Factors pair as d and `N/d`. In a perfect square the middle pair is the square root with itself, so one factor is unpaired. 36 has 1, 2, 3, 4, 6, 9, 12, 18, 36 — nine factors.

The mechanic. `72 = 2^3 * 3^2`, so the factor count is `(3+1)(2+1) = 12`. Check: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72. Twelve.

Traps

- **Forgetting 1 and N.** Both are factors of N.
- **Counting only the prime factors** when the question asked for all factors.
- **Off-by-one on counting multiples.** The count in a range is `(last - first)/step + 1`. The `+ 1` is forgotten constantly.
- **Assuming a factor of a product must divide one of the parts.** 6 divides `4 * 9 = 36` but divides neither 4 nor 9.
- **Missing the hint.** "N has an odd number of factors" is telling you N is a perfect square.
- **Negative factors.** The GRE almost always means positive factors, but read the stem.

Speed

- **Prime-factorize first, always.** Every factor question reduces to exponents once you do.
- **Add 1 to each exponent and multiply** for the factor count.
- **Odd factors:** drop all the 2s and count what is left. **Even factors:** total minus odd.
- **Stop listing at `sqrt(N)`.** Each divisor below it hands you its partner free.

2.2 Primes and prime factorization

Formulas and rules

FACT	DETAIL
1 is not prime	it has only one factor
2 is the only even prime	every other even number has 2 as a third factor
Primes under 50	2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47
Primality test	test the primes up to `sqrt(N)` only
Prime factorization	`N = p1^a1 * p2^a2 * ...`, and it is unique
Relatively prime (coprime)	greatest common divisor is 1; no shared prime factor
Composite	any integer above 1 that is not prime
0 and 1	neither prime nor composite

The test in practice. Is 191 prime? `sqrt(191)` is about 13.8, so test only 2, 3, 5, 7, 11, 13. Not even; digit sum 11 is not a multiple of 3; does not end in 0 or 5; and 7, 11, 13 all leave a remainder. **191 is prime.**

Perfect powers from the factorization. A perfect square has **all even exponents**. A perfect cube has **all exponents divisible by 3**. This turns a whole class of "smallest k such that kN is a perfect square" problems into inspection.

Traps

- **Calling 1 prime.** It is not, and this single fact drives traps constantly.
- **Assuming every prime is odd.** If a stem says "p is prime", you must consider `p = 2`.
- **Assuming every odd number is prime.** 9, 15, 21, 25 and 27 are odd and composite.
- **"Prime factors" against "factors".** 12 has six factors but only two distinct prime factors.
- **Distinct against with multiplicity.** `12 = 2^2 * 3` has three prime factors counted with multiplicity and two distinct ones. Read which is wanted.
- **Testing divisors beyond `sqrt(N)`.** Wasted time; anything above it pairs with something below.

Speed

- **Memorize the primes to 50.** It pays for itself many times over.
- **Every prime above 3 is one either side of a multiple of 6.** Useful for elimination, though not every such number is prime.
- **For "p and q are primes" questions, always test `p = 2`.** It is the exception that generates the trap answer.

2.3 Divisibility rules

The tests

DIVISOR	TEST
2	last digit is even
3	digit sum is divisible by 3
4	last **two** digits form a multiple of 4
5	last digit is 0 or 5
6	passes both the 2 test and the 3 test
8	last **three** digits form a multiple of 8
9	digit sum is divisible by 9
10	last digit is 0
11	the alternating digit sum is 0 or a multiple of 11
12	passes both the 3 test and the 4 test
25	last two digits are 00, 25, 50 or 75

The 11 test in detail. Add the digits in the odd positions from the right and subtract the digits in the even positions. For 82,918: `(8 + 9 + 8) - (2 + 1) = 22`, a multiple of 11, so 82,918 is divisible by 11.

Splitting a composite divisor — the essential caution. Break it into **coprime** factors. `12 = 3 * 4` is coprime, so testing 3 and 4 is valid. `12 = 2 * 6` is not coprime, so testing 2 and 6 is not enough: 18 passes both and is not a multiple of 12.

Divisibility of expressions. If d divides a and d divides b, then d divides `a + b`, `a - b`, and `ka` for any integer k.

Traps

- **Testing 12 as 2 and 6.** The factors must be coprime; use 3 and 4.
- **Using the 2 rule for 4.** Divisibility by 4 needs the last *two* digits.
- **Using a digit sum for 4 or 8.** The digit sum works only for 3 and 9.
- **Assuming a divisible sum means divisible terms.** 7 divides `3 + 4` and divides neither.
- **Zero.** 0 is divisible by every non-zero integer, and it is even.
- **Sign.** -12 is divisible by 3. Divisibility is about a zero remainder, and sign does not affect it.

Speed

- **Digit sum first.** It settles 3, 9 and, with the 2 test, 6 in one glance.
- **Prime-factorize the divisor and check each prime power.** For 72, check the last three digits for 8 and the digit sum for 9.

- **In "must be divisible by" questions, factor the given expression** rather than plugging in numbers.

2.4 Greatest common divisor and least common multiple

Abbreviated GCD and LCM below for compactness.

Formulas and rules

CONCEPT	RULE
GCD from the factorizations	the **lowest** power of each **shared** prime
LCM from the factorizations	the **highest** power of **every** prime present
The identity	`GCD(a,b) * LCM(a,b) = a * b` — **two numbers only**
Coprime	GCD is 1, and then the LCM is `a * b`
GCD range	`1 <= GCD <= the smaller of a and b`
LCM range	`the larger of a and b <= LCM <= a * b`
GCD divides LCM	always

The mechanic. For 84 and 120: `84 = 2^2 * 3 * 7` and `120 = 2^3 * 3 * 5`. The GCD takes the lowest power of the shared primes 2 and 3, giving `2^2 * 3 = 12`. The LCM takes the highest power of every prime, giving `2^3 * 3 * 5 * 7 = 840`. Check: `12 * 840 = 10,080` and `84 * 120 = 10,080`.

Traps

- **Swapping the two.** The GCD is small, at most the smaller number; the LCM is large, at least the larger. If your GCD exceeds either number, you computed the LCM.
- **Using the highest power for the GCD.** It takes the *lowest* power, and only of *shared* primes.
- **Applying the identity to three numbers.** It fails: for 2, 4 and 8 the GCD is 2 and the LCM is 8, but the product of the three is 64.
- **"Divisible by 4 and by 6" read as "divisible by 24".** It means divisible by 12, their LCM.
- **Forgetting a prime that appears in only one number** when building the LCM.
- **Word-problem misread.** "The same number of items in every group" is a GCD question; "the smallest length both can be cut into" is an LCM question.

Speed

- **Lay the primes down a column** with exponents across. The GCD is the row minimum, the LCM the row maximum.
- **Repeated remainders for two large numbers.** Replace the larger number by the remainder it leaves on division by the smaller, and repeat until the remainder is 0; the last non-zero value is the GCD. For 252 and 105: 252 leaves 42, then 105 leaves 21, then 42 leaves 0, so the GCD is 21.
- **If one number divides the other,** the GCD is the smaller and the LCM is the larger. Instant.
- **Consecutive integers are always coprime,** so the GCD is 1 and the LCM is the product.

2.5 Consecutive integers

Formulas and rules

CONCEPT	RULE
n consecutive integers	`n, n+1, n+2, ...`
n consecutive even or odd integers	`n, n+2, n+4, ...` — the gap is **2**
Sum of an evenly spaced set	`count * average`
Average of an evenly spaced set	`(first + last)/2`
Count of integers from a to b	`b - a + 1`
Product of k consecutive integers	always divisible by `k!`
Among any 2 consecutive	one is even
Among any 3 consecutive	one is divisible by 3
Sum of the first n positive integers	`n(n+1)/2`
Sum of the first n odd numbers	`n^2`

Product divisibility. Two consecutive integers give a multiple of `2! = 2`; three give a multiple of `3! = 6`; four give a multiple of `4! = 24`. So `n(n+1)(n+2)` is a multiple of 6 for every integer n.

Sum parity. The sum of an **odd** number of consecutive integers is divisible by that count — three consecutive integers sum to three times the middle one. The sum of an **even** number of consecutive integers is **never** divisible by the count.

Setting up. For three consecutive integers, `n-1, n, n+1` beats `n, n+1, n+2`, because the sum collapses to `3n` immediately.

Traps

- **Assuming consecutive means positive.** -3, -2, -1, 0, 1 are consecutive integers.
- **Spacing for consecutive even or odd integers.** The gap is 2. Writing `n, n+1, n+2` for consecutive odd integers is the standard error.
- **Count off by one.** From 7 to 19 inclusive there are `19 - 7 + 1 = 13` integers.
- **Expecting the average to be a member.** The average of 4, 5, 6, 7 is 5.5.
- **Assuming the sum is always divisible by the count.** True for odd counts only.

Speed

- **Center the variable** on an odd-sized set and the sum collapses at once.
- **`Sum = count * average`, with `average = (first + last)/2`.** The most reusable formula in the topic, and it works for any evenly spaced set, not just consecutive integers.
- **For "must be divisible by" questions, count the consecutive terms in the product** and apply the `k!` rule.
- **Test `n = 0`, `n = 1` and `n = 2`** on any must-be-true statement.

2.6 Even, odd and parity

The table

OPERATION	RESULT
even + even	even
odd + odd	**even**
even + odd	odd
even - even	even
odd - odd	**even**
even * even	even
even * odd	even
odd * odd	**odd**
`even^n` for `n >= 1`	even
`odd^n`	odd

The two facts students forget: `odd + odd = even`, and `even * anything = even`. The only way to get an odd product is for **every** factor to be odd.

Division has no parity rule. `8/4 = 2`, `6/2 = 3`, `8/16 = 0.5`. Never reason about parity through division.

Special cases. 0 is even, because it is `2 * 0`. Negative integers have parity too: -6 is even and -7 is odd. Consecutive integers alternate parity.

Traps

- **`odd + odd = odd`.** It is even. The most common parity slip on the test.
- **Forgetting 0 is even.** "How many even integers lie between -3 and 3" must count -2, 0 and 2.
- **Applying parity to division.** It does not apply.
- **Assuming a variable is an integer.** Parity is defined only for integers; if the stem does not say integer, parity reasoning may not apply at all.
- **`even^0`.** `2^0 = 1`, which is odd. The rule needs `n >= 1`.
- **Reading "not even" as "positive odd".** Negative odd integers exist.

Speed

- **Substitute 2 for even and 3 for odd,** then test 0 and 1 separately as edge cases.
- **Throw away every even coefficient** — it forces its term even, so it cannot change the parity of a sum.
- **`n(n+1)` is always even.** This identity resolves many must-be-true questions instantly.
- **On quantitative comparison, test both parities** of any unrestricted integer variable.

2.7 Digits and place value

Formulas and rules

CONCEPT	RULE
Two-digit number	`10a + b`, with a from 1 to 9 and b from 0 to 9
Three-digit number	`100a + 10b + c`
Reversing a two-digit number	`10b + a`
Number minus its reverse	`9(a - b)` — always a multiple of **9**
Number plus its reverse	`11(a + b)` — always a multiple of **11**
Leading digit	cannot be 0
Digit sum	tests divisibility by 3 and 9
Units digit of a product	depends only on the units digits of the factors

Counting with digit constraints — the slot method. Draw one blank per digit, count the allowed choices for each, multiply, and handle the leading slot first because it excludes 0. How many three-digit numbers have all distinct digits? `9 * 9 * 8 = 648`.

Traps

- **Writing `ab` for a two-digit number.** That means a times b. You need `10a + b`.
- **Allowing 0 as a leading digit.**
- **Confusing digits with numbers.** "The digits sum to 12" is a very different constraint from "the number is 12".
- **Assuming the digits are distinct** when the stem does not say so, or the reverse.
- **Decimal place value.** In 3.147 the 4 is in the hundredths place — tenths, hundredths, thousandths, going right.
- **Forgetting the digit range.** Any solution with a digit outside 0 to 9 is invalid, which often rules out an otherwise clean algebraic answer.

Speed

- **Expand immediately to `10a + b`.** Never work with the digits informally.
- **"27 less when reversed" instantly gives `a - b = 3`,** from the `9(a - b)` identity.
- **Units-digit questions need only the units digits.** Discard everything else.
- **For divisibility by 9, use the digit sum rather than dividing.**

2.8 Number families, closure and the zero-one edge cases

The families, nested

SET	MEMBERS
Counting or natural numbers	1, 2, 3, ...
Whole numbers	0, 1, 2, 3, ...
Integers	..., -2, -1, 0, 1, 2, ...
Rational numbers	any `a/b` with a and b **integers** and b not 0
Irrational numbers	`sqrt(2)`, `pi` — non-terminating **and** non-repeating decimals
Real numbers	everything on the number line

Closure. The integers are closed under addition, subtraction and multiplication but **not** division. The rationals are closed under all four, with a non-zero divisor. The irrationals are closed under **none** of them: `sqrt(2) * sqrt(2) = 2` and `sqrt(2) + (-sqrt(2)) = 0`. But two rules always hold: **rational plus irrational is irrational**, and **non-zero rational times irrational is irrational**.

Squares and cubes

PROPERTY	SQUARES	CUBES
Sign	never negative	keeps the sign of the base
Possible units digits	0, 1, 4, 5, 6, 9	any digit
Factor count	odd	not necessarily
Prime exponents	all even	all multiples of 3

A perfect square never ends in 2, 3, 7 or 8. A fast elimination tool.

Zero and one. 0 is an integer, is even, is neither positive nor negative, and is a multiple of every integer. `0/5 = 0` is fine; `5/0` is undefined. 1 is neither prime nor composite, and `1^n = 1`.

The two ordering chains worth memorizing

```
If 0 < x < 1:   x^2  <  x  <  sqrt(x)  <  1/x
If x > 1:       1/x  <  sqrt(x)  <  x  <  x^2
```

Traps

- **"Number" does not mean "integer".** Unless the stem says integer, x could be 2.5 or -1/3.
- **Assuming positive.** "x is a number" allows negatives and zero.
- **Assuming irrational times irrational is irrational.** `sqrt(2) * sqrt(2) = 2`.
- **Assuming every non-terminating decimal is irrational.** `0.333...` is `1/3`. Irrational means non-terminating *and* non-repeating.
- **Assuming `x^2 > x`.** False for `0 < x < 1`, and for `x = 0` and `x = 1`.
- **Reciprocal ordering.** For positives, reciprocals reverse the order: `2 < 3` but `1/2 > 1/3`.

Speed

- **Run the FROZEN check before answering any must-be-true question.** If a fraction, zero or a negative breaks the statement, it is not must-be-true.
- **Units digits eliminate perfect-square candidates in one glance.**
- **An odd count of factors means a perfect square** — a common disguised hint.

Section 3 — Algebra

3.1 The identities you must know cold

IDENTITY	EXPANSION
`(a + b)^2`	`a^2 + 2ab + b^2`
`(a - b)^2`	`a^2 - 2ab + b^2`
`(a + b)(a - b)`	`a^2 - b^2`
`(a + b)^3`	`a^3 + 3a^2 b + 3ab^2 + b^3`
`a^3 + b^3`	`(a + b)(a^2 - ab + b^2)`
`a^3 - b^3`	`(a - b)(a^2 + ab + b^2)`
`(x + a)(x + b)`	`x^2 + (a + b)x + ab`

The derived forms the GRE keeps reaching for

```
a^2 + b^2 = (a + b)^2 - 2ab
a^2 + b^2 = (a - b)^2 + 2ab
(a + b)^2 - (a - b)^2 = 4ab
If x + 1/x = k, then x^2 + 1/x^2 = k^2 - 2
```

When a stem gives you `a + b` and `ab` rather than a and b, it is an identity question. You will almost never need a and b separately.

Order of operations. Brackets, exponents, multiply and divide left to right, then add and subtract left to right. So `2 * 3^2 = 18`, not 36.

Like terms combine; unlike terms do not. `3x^2 + 5x^2 = 8x^2`, but `3x^2 + 5x` stays as it is.

Complex fractions. Multiply the top and the bottom by the least common multiple of the inner denominators.

Traps

- **`(a + b)^2 = a^2 + b^2`.** The `2ab` is missing. This is the single most common algebra error on the test.
- **`(a - b)^2 = a^2 - b^2`.** No. The correct expansion is `a^2 - 2ab + b^2`; `a^2 - b^2` is the difference of squares, which factors as `(a + b)(a - b)`.
- **Distributing a negative to only the first term.** `-(x - 3) = -x + 3`.
- **Canceling across a plus sign.** `(x + 3)/3` is not x. You may cancel common **factors**, never terms.
- **`sqrt(a^2 + b^2) = a + b`.** False.
- **Dividing by a variable that might be 0.** From `x^2 = 3x`, factoring gives `x(x - 3) = 0`, so `x = 0` or `x = 3`. Dividing by x loses the root `x = 0`.

Speed

- **Look for `a^2 - b^2` in any numeric subtraction of squares.** It turns heavy arithmetic into a two-step multiplication.
- **Factor before you expand.** GRE answer choices are usually in factored form.
- **For any `x + 1/x` problem, square both sides.** That single move solves nearly all of them.
- **Check an identity by substituting `x = 2`** if you are unsure.

3.2 Factoring

The checklist, in this order

1. **Common factor first. Always.** `3x^2 + 6x = 3x(x + 2)`.
2. **Two terms?** Look for a difference of squares, or a sum or difference of cubes.
3. **Three terms?** Try the `x^2 + (p+q)x + pq` pattern, or a perfect-square trinomial.
4. **Four terms?** Group in pairs.

PATTERN	FACTORED FORM
a^2 - b^2	(a + b)(a - b)
a^2 + 2ab + b^2	(a + b)^2
a^2 - 2ab + b^2	(a - b)^2
a^3 + b^3	(a + b)(a^2 - ab + b^2)
a^3 - b^3	(a - b)(a^2 + ab + b^2)
x^2 + (p+q)x + pq	(x + p)(x + q)

a^2 + b^2 does not factor over the real numbers.

A leading coefficient. For ax^2 + bx + c, find two numbers whose product is ac and whose sum is b, then split the middle term. For 6x^2 + 11x + 3: ac = 18, and 9 and 2 multiply to 18 and sum to 11, so 6x^2 + 9x + 2x + 3 = 3x(2x + 3) + (2x + 3) = (3x + 1)(2x + 3).

Grouping. xy + 3x + 2y + 6 = x(y + 3) + 2(y + 3) = (x + 2)(y + 3).

The zero-product property. AB = 0 means A = 0 or B = 0. This is why you factor rather than divide.

Traps

- **Forgetting the common factor.** 2x^3 - 18x written as (x - 3)(x + 3) has lost the 2x.
- **Dividing instead of factoring.** From x^2 = 5x, dividing by x loses x = 0.
- **Trying to factor a^2 + b^2.**
- **Sign errors in the trinomial split.** Expand to check; it costs five seconds.
- **Canceling terms.** In (x^2 - 4)/(x + 3) nothing cancels, because x + 3 is not a factor of the numerator.
- **Ignoring domain restrictions.** After canceling (x + 2), the original expression is still undefined at x = -2.

Speed

- **Any (something)^2 - (something)^2 factors instantly,** even with variables inside: (x+1)^2 - 9 = (x + 4)(x - 2).
- **If the choices are factored, expand one and compare.**
- **Sign shortcut for x^2 + bx + c:** when c is positive, both factors share the sign of b; when c is negative, the factors have opposite signs and the larger carries the sign of b.
- **Check any factorization at x = 1.** Both forms must agree.

3.3 Linear equations in one variable

The procedure

STEP	RULE
Clear fractions	multiply both sides by the least common multiple of the denominators
Clear brackets	distribute, watching signs
Collect	variables one side, constants the other
Isolate	divide by the coefficient
Check	substitute back

How many solutions

AFTER SIMPLIFYING	SOLUTIONS
x = a specific number	exactly one
0 = 0	infinitely many
0 = 5	none

Literal equations use identical steps. To solve A = P(1 + rt) for r: A/P = 1 + rt, so r = (A - P)/(Pt). Treat every other letter as a number.

The translation table

ENGLISH	ALGEBRA
is, was, will be	`=`
more than, sum, increased by	`+`
less than, difference, decreased by	`-` — **mind the order**
of, product, times	`*`
per, out of, ratio	`/`
what, a number	the variable

"5 less than x" is `x - 5`, not `5 - x`.

Traps

- **Distributing a negative over only the first term.**
- **"5 less than x" written as `5 - x`.**
- **Multiplying both sides by an expression containing the variable** without noting it could be zero. That can create or destroy solutions.
- **Answering the wrong quantity.** The stem may want `2x + 1`, not x.
- **Assuming a solution exists.** Some questions are testing whether you notice "no solution".

Speed

- **Clear fractions first.** Integers are always faster than fractions.
- **Move the smaller variable term** to avoid negatives.
- **Back-solve from the choices** when the algebra is ugly; start with the middle value so one test tells you which way to move.
- **Check by substitution** on anything that took over a minute. Ten seconds, and it catches sign errors.

3.4 Systems of equations

Methods

METHOD	PROCEDURE
Substitution	isolate one variable, substitute into the other equation
Elimination	scale one or both equations so a variable cancels on adding or subtracting
The shortcut	add or subtract the equations as they stand and see what emerges

How many solutions — the geometric reading

SITUATION	THE LINES	SOLUTIONS
Different slopes	cross once	exactly one
Same slope, different intercept	parallel	none
Same slope, same intercept	identical	infinitely many

Algebraically, for `a1 x + b1 y = c1` and `a2 x + b2 y = c2`: one solution when `a1/a2` differs from `b1/b2`; no solution when `a1/a2 = b1/b2` but neither equals `c1/c2`; infinitely many when `a1/a2 = b1/b2 = c1/c2`.

Three variables need three independent equations. Eliminate one variable twice to get down to a two-by-two system.

Traps

- **Solving for both variables when only the sum is wanted.** Forty wasted seconds.
- **Sign errors in elimination.** Subtracting an equation changes the sign of **every** term in it.
- **Confusing "none" with "infinitely many".** Same slope and a different intercept gives none; same slope and the same intercept gives infinitely many.
- **Assuming two equations always pin down two unknowns.** Not if one is a multiple of the other.
- **Answering x when the question asked for y, or for `xy`, or for `x/y`.**

Speed

- **Ask what is actually wanted before you solve.** If it is `x + y` or `x - y`, try adding or subtracting the raw equations.

- **Symmetric coefficients are a signal.** For `ax + by = m` and `bx + ay = n`, adding gives `(a+b)(x+y) = m+n` and subtracting gives `(a-b)(x-y) = m-n`. Both in one step.
- **Elimination beats substitution** when no coefficient is 1.
- **Back-substitute into the equation you did not use.** A free check.
- **For "no solution" questions, compare the coefficient ratios** rather than solving.

3.5 Inequalities

Formulas and rules

OPERATION	EFFECT ON THE SIGN
Add or subtract anything	unchanged
Multiply or divide by a positive	unchanged
Multiply or divide by a **negative**	**reversed**
Take reciprocals of two positives	reversed: `2 < 3` but `1/2 > 1/3`
Square both sides	safe only when both sides are non-negative

Compound inequalities. `a < x < b` means **both** conditions, an interval. `x < a or x > b` means **either**, two rays. Operate on all three parts at once: from `-3 < 2x + 1 < 7`, subtract 1 throughout and divide by 2, giving `-2 < x < 3`.

Combining inequalities. You may **add** two that point the same way: from `a > b` and `c > d` you get `a + c > b + d`. You may **not subtract** them. You may multiply only when every quantity is positive.

Quadratic inequalities — the critical-point method. For `x^2 - 5x + 6 > 0`, factor to `(x-2)(x-3) > 0`. The critical points are 2 and 3. Test one value in each region: below 2 the product is positive and works; between 2 and 3 it is negative and fails; above 3 it is positive and works. So the answer is `x < 2 or x > 3`. For `< 0` the answer is the middle region, `2 < x < 3`.

Traps

- **Forgetting to flip when dividing by a negative.** The most common inequality error, full stop.
- **Dividing by a variable of unknown sign.** From `ax > a` you cannot divide by a.
- **Subtracting inequalities.** Flip the second and add instead.
- **`x^2 < 9` giving `x < 3`.** It gives `-3 < x < 3`.
- **Squaring carelessly.** `-5 < 2` is true; `25 < 4` is false.
- **Treating strict endpoints as included.** Watch `<` against `<=`.

Speed

- **For range questions, test the four corner combinations** of the extremes and take the widest result. It works for addition, subtraction and multiplication — and for division only when the divisor's range excludes 0, since a divisor straddling 0 makes the quotient unbounded.
- **Draw the number line and shade.** It prevents every and-or mix-up.
- **On quantitative comparison, hunt for a counterexample** rather than trying to prove equality.
- **Keep everything positive** by moving terms instead of multiplying by negatives. That sidesteps the flip rule entirely.

3.6 Quadratic equations

Formulas and rules

CONCEPT	FORMULA
Standard form	`ax^2 + bx + c = 0`
The quadratic formula	`x = [-b +/- sqrt(b^2 - 4ac)] / (2a)`
Discriminant	`D = b^2 - 4ac`
Sum of the roots	`-b/a`
Product of the roots	`c/a`
Zero-product property	`AB = 0` means `A = 0` or `B = 0`
Equation from its roots r and s	`x^2 - (r + s)x + rs = 0`

The discriminant counts the real roots

D	ROOTS
`D > 0`	two distinct real roots
`D = 0`	exactly one real root, repeated
`D < 0`	no real roots

Sum and product in practice. For `2x^2 - 10x + 8 = 0`, the sum of the roots is `10/2 = 5` and the product is `8/2 = 4`. The roots are 1 and 4, which confirms both.

Completing the square. `x^2 + 6x + 5 = 0` becomes `(x + 3)^2 - 9 + 5 = 0`, so `(x + 3)^2 = 4`, giving `x = -1` or `x = -5`.

Quadratic in disguise. `x^4 - 5x^2 + 4 = 0` is quadratic in `u = x^2`: `u^2 - 5u + 4 = 0` gives `u = 1` or `u = 4`, so `x = +/-1` or `x = +/-2`. **Four** solutions.

Traps

- **Missing the negative root.** From `x^2 = 49`, x is 7 **or** -7. When the GRE hands you `x^2 = k`, it is usually testing exactly this.
- **Sum of roots written as `b/a`.** It is `-b/a`. The minus sign is forgotten constantly.
- **Dividing by x** and losing the root `x = 0`.
- **Assuming two roots always exist.** With `D < 0` there are none over the reals.
- **Forgetting to set the equation to zero** before factoring. `x^2 = 5x + 24` must become `x^2 - 5x - 24 = 0`.
- **Extraneous roots.** After squaring both sides of an equation with a square root in it, always substitute back; some roots will be invalid.

Speed

- **Try factoring first.** GRE quadratics are built to factor most of the time.
- **When the question asks for the sum or the product of the roots, use `-b/a` and `c/a`** and never solve for the roots individually.
- **Discriminant questions never need the roots.** Just compute `b^2 - 4ac`.
- **Recognize the perfect-square trinomials** `x^2 + 2kx + k^2` and `x^2 - 2kx + k^2`.
- **Back-solve from the choices** when a quadratic looks ugly.

3.7 Functions

Formulas and rules

CONCEPT	MEANING
`f(x)`	the output for input x — a machine, **not** a product
`f(3)`	substitute 3 for every x
`f(a + 1)`	substitute `(a + 1)` for every x, keeping the brackets
Domain	all allowed inputs
Range	all possible outputs
`f(g(x))`	apply g **first**, then f
`f^(-1)(x)`	swap x and y, then solve for y
Vertical line test	a graph is a function if no vertical line meets it twice

The only two domain restrictions the GRE tests: a denominator cannot be zero, and the expression under an even root cannot be negative. For `f(x) = sqrt(x - 3)/(x - 5)`, the domain is `x >= 3` with x not equal to 5.

Composites do not commute. `f(g(x))` is usually different from `g(f(x))`.

Strange-symbol functions. The test often invents an operation, such as `a # b = a^2 - 2b`. These are ordinary functions in unusual clothing. Substitute exactly as defined and do not overthink it.

Traps

- **Reading `f(x)` as f times x.**

- **`f(a + b) = f(a) + f(b).`** Almost never true. For `f(x) = x^2`, `f(3) = 9` but `f(1) + f(2) = 5`.
- **Order in composites.** `f(g(x))` applies the **inner** function first.
- **Dropping brackets when substituting.** For `f(x) = x^2`, `f(-3) = 9`, not -9.
- **Forgetting domain restrictions,** especially the zero denominator, which often eliminates a choice.
- **`2f(x)` against `f(2x).`** One doubles the output, the other doubles the input.

Speed

- **Work from the inside out** on composites.
- **Cover the x with your finger.** Whatever is in the brackets goes wherever your finger was.
- **For "which function has this graph", test one easy point** such as `x = 0` and eliminate.
- **Inverse check:** `f(f^(-1)(x))` must give back x. A ten-second verification.

3.8 Sequences and series

Arithmetic sequence, first term a and common difference d

QUANTITY	FORMULA
nth term	`a_n = a + (n - 1)d`
Sum of the first n terms	`S_n = (n/2)(first + last) = (n/2)[2a + (n-1)d]`
Average of the terms	`(first + last)/2`
Number of terms from a to L	`(L - a)/d + 1`

Geometric sequence, first term a and common ratio r

QUANTITY	FORMULA
nth term	`a_n = a * r^(n-1)`
Sum of the first n terms	`S_n = a(r^n - 1)/(r - 1)` for r not 1
Infinite sum, only when `|r| < 1`	`S = a/(1 - r)`

Summation results worth knowing

```
1 + 2 + ... + n          = n(n+1)/2
first n odd numbers      = n^2
first n even numbers     = n(n+1)
```

Recursive sequences. Given `a_1 = 3` and `a_(n+1) = 2 a_n - 1`, just generate: 3, 5, 9, 17, 33. GRE recursive questions rarely run past six or eight terms, so computing beats finding a closed form.

Traps

- **Off by one in the nth term.** It is `a + (n-1)d`. The first term uses no d at all.
- **Off by one in counting terms.** From 24 to 198 in steps of 6 there are 30 terms, not 29.
- **Confusing arithmetic with geometric.** Check differences *and* ratios before choosing a formula.
- **Using the infinite-sum formula when `|r| >= 1`.** There is no finite sum.
- **Assuming a pattern from three terms.** 1, 2, 4 could continue 8 or 7. Look for a stated rule.
- **Sum against term.** "The 10th term" and "the sum of the first 10 terms" are different questions.

Speed

- **`Sum = count * average`, with `average = (first + last)/2.`** One line that replaces both sum formulas for any evenly spaced set.
- **Count terms with `(last - first)/step + 1.`** Say the "plus one" out loud every time.
- **For recursive sequences, list the terms.** Almost always faster than algebra.
- **Look for telescoping.** Sums like `1/(1*2) + 1/(2*3) + ...` collapse, because each term is `1/n - 1/(n+1)`.

3.9 Algebraic exponents and radicals

Formulas and rules

RULE	STATEMENT
`x^a * x^b`	`x^(a+b)`
`x^a / x^b`	`x^(a-b)`
`(x^a)^b`	`x^(ab)`
`(xy)^a`	`x^a y^a`
`x^(-a)`	`1/x^a`
`x^(a/b)`	the b-th root of `x^a` — **the denominator is the root**
If `x^a = x^b`	then `a = b`, **provided** x is not 0, 1 or -1

The equating rule has conditions. If `x = 1` then `1^5 = 1^9` without 5 equaling 9.

Radical rules

```
sqrt(xy)  = sqrt(x) * sqrt(y)        for x, y >= 0
sqrt(x/y) = sqrt(x)/sqrt(y)          for x >= 0 and y > 0
sqrt(x^2) = |x|                      NOT x
sqrt(x) + sqrt(y) is NOT sqrt(x + y)
```

Rationalizing. `1/sqrt(x) = sqrt(x)/x`. For `1/(a + sqrt(b))`, multiply above and below by the conjugate `a - sqrt(b)` to get `(a - sqrt(b))/(a^2 - b)`.

Factoring with exponents — the highest-yield skill in the topic. `2^(n+2) + 2^n = 2^n(2^2 + 1) = 5 * 2^n`.

Traps

- **Equating exponents when the base could be 0, 1 or -1.**
- **`x^a + x^b = x^(a+b)`.** No such rule. Factor instead.
- **`sqrt(x^2) = x`.** It is `|x|`.
- **Negative exponents read as negative numbers.** `x^(-2) = 1/x^2`, positive for any non-zero real x.
- **Flipping a fractional exponent.** `x^(2/3)` is the cube root of `x^2`, not `x^(3/2)`.
- **Canceling a variable base that could be zero.**

Speed

- **Rewrite every base as a prime power.** Nearly all exponent equations become linear in the exponent.
- **Factor out the smallest power** on any sum or difference of powers with the same base.
- **When the answer must be independent of n, factoring is the intended route.**
- **Conjugates clear any two-term radical denominator in one step.**
- **Pull out perfect squares first.** `sqrt(48) = 4 sqrt(3)` before anything else.
- **Check with a number.** Substitute `x = 4` into your simplified form and the original; they must agree.

Section 4 — Word Problems

4.1 Translating a word problem into algebra

The translation dictionary

ENGLISH	ALGEBRA
is, are, was, will be, gives	`=`
sum, more than, increased by, together	`+`
difference, less than, decreased by, fewer	`-`
of, product, times, twice, triple	`*`
per, quotient, ratio, out of, divided by	`/`
what, a number, how many	the variable
consecutive integers	`n, n+1, n+2`
percent	`/100`

The two order traps. "5 less than x" is `x - 5`. "The quotient of a and b" is `a/b`, in the order given. So "7 less than twice a number" is `2n - 7`.

Choose the variable well. Name the **smallest or simplest** quantity x. If Ram has three times as much as Sita, let Sita be x so that Ram is `3x`, and you avoid fractions entirely.

The setup templates

SITUATION	EQUATION
Total	part + part = whole
Comparison	bigger = smaller + difference
Rate	`rate * time = amount`
Before and after	new = old plus or minus the change
Two conditions	write two equations

Traps

- **Solving for x but answering a different quantity.** Circle what is asked before you start.
- **Ignoring units.** Mixing minutes with hours silently ruins the equation.
- **Assuming the answer must be a whole number.** People and tickets must be; money and time need not be.
- **Too many variables.** Two unknowns need two equations. With only one relationship, express everything through a single variable.
- **Not using all the information.** If a number in the stem never appears in your equations, you have probably misread something.

Speed

- **Write "Let x =" in full words, with units.** Four seconds, and it prevents the wrong-quantity error.
- **Translate sentence by sentence.** Do not build the whole equation in your head.
- **Back-solve from the choices** when the setup is messy, starting from the middle value.
- **Pick smart numbers** when the problem is all percents or fractions with no amounts: use 100, or the least common multiple of the denominators.
- **Sanity-check the size of the answer.** A discount problem whose answer exceeds the original price has a sign error in it.

4.2 Distance, speed and time

Formulas and rules

QUANTITY	FORMULA
Distance	`Speed * Time`
Speed	`Distance / Time`
Time	`Distance / Speed`
Average speed	`Total distance / Total time`
Same **distance** at speeds a and b	`2ab/(a + b)`
Same **time** at speeds a and b	`(a + b)/2`
Approaching each other	the gap closes at `s1 + s2`
Chasing, same direction	the gap closes at `s1 - s2`

Always draw the table. One row per traveler, three columns for distance, speed and time. Fill in what you know, express the rest through one variable, and use the shared quantity to write the equation.

```
          Distance   Speed   Time
   A
   B
```

Relative motion. Time to meet is `gap / (s1 + s2)`. Time to catch is `gap / (s1 - s2)`.

Unit consistency. Speed in kilometers per hour needs time in hours: 45 minutes is 0.75 hours.

Traps

- **Averaging the two speeds.** Valid only when the *times* are equal, never when the distances are.
- **Unit mismatch.** Minutes with kilometers per hour, or meters with kilometers.
- **Adding times for travelers who move simultaneously.** They share the clock.
- **Forgetting the head start** in a catch-up problem.
- **"Speed increased by 25 percent".** The multiplier is 1.25, so the time becomes `1/1.25 = 0.8` of the original, a 20 percent reduction, not 25.
- **Round trip.** It is *twice* the one-way distance, and some stems give one and ask for the other.
- **Stops and breaks.** A 20-minute halt still counts in the average speed for the whole trip.

Speed

- **Pick a convenient distance** when none is given, usually the least common multiple of the speeds. For 60 and 90, take 180 kilometers each way: the times are 3 hours and 2 hours, so 360 kilometers in 5 hours, giving 72 kilometers per hour. No formula needed.
- **Over equal distances, the average speed always lands nearer the slower speed.** Predict that before computing. (Over equal *times* it is exactly the midpoint.)
- **For meeting problems, add the speeds and divide the gap.** One line.

4.3 Work and rate

Formulas and rules

CONCEPT	FORMULA
Individual rate	`1 job / time taken`
Working together	`R_A + R_B`
Fill against drain	`R_fill - R_drain`
Two workers taking a and b hours	together `ab/(a + b)` hours
Work done in time t	`Rate * t`
Remaining work	`1 - (work already done)`
Worker-days	`workers * days` is conserved, at equal efficiency

Two methods, and which to use.

Fractional rates. A takes 6 hours, so the rate is `1/6` of the job per hour. B takes 3 hours, so `1/3`. Together `1/6 + 1/3 = 1/2`, giving 2 hours.

Least-common-multiple units, usually faster. Let the job be `LCM(6, 3) = 6` units. A does 1 unit an hour, B does 2, together 3, so `6/3 = 2` hours. **No fractions at all.**

Worker-days. If 6 workers take 10 days, the job is 60 worker-days, so 15 workers take `60/15 = 4` days.

Traps

- **Averaging the times.** A at 6 hours and B at 3 hours do not take 4.5 hours together. The answer must be **less than the faster worker's time**.
- **Adding times.** Never.
- **Forgetting the drain is negative.** Subtract it.
- **Using worker-days when efficiencies differ.** It assumes identical workers.
- **Skipping the sanity check.** The combined time is always shorter than any individual time.
- **"Half the job" questions.** Find the rate first, then apply it to half a job.

Speed

- **Use least-common-multiple units.** It removes every fraction, and it is worth learning before anything else in this topic.
- **Two-worker shortcut `ab/(a+b)`.** For 12 and 18: `216/30 = 7.2` hours. Instant.
- **Check the bound.** The combined time always lies between half the faster time and the faster time.
- **Worker-days is conserved,** so more workers means proportionally fewer days.

4.4 Mixtures and solutions

The one habit that solves half the topic: for every solution mentioned, write down `concentration * volume` — the amount of the pure thing — before doing anything else.

Formulas and rules

CONCEPT	RULE
Amount of pure component	`concentration * volume`
The balance equation	`c1 V1 + c2 V2 = c_final * (V1 + V2)`
Water	0 percent
Pure solute	100 percent
Adding water	the denominator rises, the numerator does not
Adding pure solute	both rise
Evaporation	the denominator falls, the numerator does not
Alligation ratio	the two volumes are in the ratio of the **opposite** gaps to the target
Repeated replacement, n times	the remaining fraction is `(1 - f)^n`

Alligation. To mix 20 percent and 50 percent solutions to reach 30 percent: the gaps to the target are 10 and 20, so the volumes are in the ratio `20 : 10`, which is `2 : 1` — **more of the source nearer the target.**

The six traps

TRAP	WRONG	RIGHT
Averaging the concentrations	15 and 40 percent mixed gives 27.5 percent	weight by volume
Total left unchanged when adding	`10/40` after adding 10 liters of water	`10/(40+10)` = 20 percent
Pure solute treated as a solution	`(3+x)/15 = 0.5`	`(3+x)/(15+x) = 0.5`
Alligation ratio inverted	20 and 50 to reach 30 gives `1:2`	`2:1`
Percent used as a whole number	`25 * 40` liters of acid	`0.25 * 40` = 10 liters
Removal treated as removing pure solute	the second scoop removes 8 liters of milk	it removes `8 *` the current milk fraction

Speed

- **Ask "did the total change?" after every operation.** That one question defeats three of the six traps.

- **Set the total to 100** when the problem gives no volumes. The answer then reads off directly.
- **State whether the answer is above or below the midpoint before computing.** It kills the plain-average trap instantly.
- **Work in liters of the pure substance, not in percentages,** until the final step.
- **For repeated replacement, use `(1 - f)^n`** rather than stepping through each round.
- **Check that the pure component still balances** at the end. It catches almost every arithmetic slip.

4.5 Simple and compound interest

Throughout, r is the rate as a **decimal** (8 percent is 0.08) and t is in years, matching the rate's period.

Formulas and rules

CONCEPT	FORMULA
Simple interest	`SI = P * r * t`
Amount, simple	`A = P(1 + rt)`
Amount, compound	`A = P(1 + r)^t`
Compound interest	`CI = P[(1 + r)^t - 1]`
Compounded n times a year	`A = P(1 + r/n)^(nt)`
The 2-year difference	`CI - SI = P * r^2`
Doubling time, roughly	about `72 / (rate as a percent)` years

Compounding frequency. For 12 percent a year compounded quarterly over 2 years: the rate per quarter is 3 percent and there are 8 periods, so `A = P(1.03)^8`.

The 2-year difference is worth memorizing. For `P = 10,000` at 10 percent, the difference is `10,000 * 0.01 = 100`.

Traps

- **Using the annual rate with a non-annual period.** Quarterly compounding needs `r/4` and `4t`.
- **Forgetting to subtract the principal.** "Interest earned" is `A - P`, not A.
- **Treating the percentage as a whole number.** 8 percent is 0.08 in the formula.
- **Assuming they always differ.** Simple and compound interest are **equal for exactly one compounding period** and diverge after that — a standard quantitative comparison setup.
- **Missing the word "compounded".** If it is absent, the interest is usually simple.
- **"6 percent a year compounded semi-annually"** is 3 percent per half-year.

Speed

- **Compute the multiplier, then raise it to the power.** Keep `(1.08)^3` unevaluated until the end.
- **For 2-year problems, use `P r^2`.** Five seconds.
- **Year by year beats the formula for two or three years:** add 10 percent, then 10 percent of the new total.
- **On comparison questions, compound beats simple for `t > 1`** — you may not need to compute at all.

4.6 Profit, loss and discount

CP is cost price, SP is selling price and MP is the marked or list price.

Formulas and rules

CONCEPT	FORMULA
Profit	`SP - CP`
Profit percent	`(SP - CP)/CP * 100` — the base is always **CP**
Loss percent	`(CP - SP)/CP * 100`
SP from a profit percent	`SP = CP(1 + p/100)`
CP from SP and a profit percent	`CP = SP / (1 + p/100)`
Discount percent	`(MP - SP)/MP * 100` — the base is always **MP**
SP after a discount	`SP = MP(1 - d/100)`
Markup percent	`(MP - CP)/CP * 100`
The whole chain	`SP = CP * (1 + markup) * (1 - discount)`

Successive discounts multiply. Two 20 percent discounts give `0.8 * 0.8 = 0.64`, a **36 percent** total discount, not 40.

Traps

- **Computing profit percent on SP.** The base is CP.
- **Computing discount percent on CP.** The base is MP.
- **Adding successive discounts.** 20 percent then 10 percent is 28 percent, not 30.
- **Assuming an equal profit and loss percentage cancel.** The result is always a net **loss**.
- **Confusing markup with profit.** A 50 percent markup is not 50 percent profit once a discount follows.
- **Applying the discount before the markup.** Follow the order in the sentence.

Speed

- **Set CP to 100** when the problem is all percentages.
- **Use multipliers for the whole chain** in one line.
- **For reverse problems, divide by the multiplier** — never subtract the same percent.
- **Equal profit and loss at x percent gives a net loss of `(x/10)^2` percent.** At 10 percent each way, a 1 percent loss.

4.7 Age problems

Formulas and rules

PHRASE	ALGEBRA, WITH PRESENT AGE X
t years ago	`x - t`
in t years	`x + t`
twice as old as	`2 *` the other
older than by k	`x = y + k`
the age difference	**constant** over time
sum S and difference D	the ages are `(S+D)/2` and `(S-D)/2`

The table method. Build three columns and fill in what is known, then write one equation per condition.

```
Person    t years ago    Now    In t years
  A          a - t        a       a + t
  B          b - t        b       b + t
```

Traps

- **Adding t to only one person.** Everyone ages together.
- **"Was twice as old" applied to present ages.** That condition belongs in the past column.
- **Assuming the age ratio is constant.** Only the difference is.
- **Answering the wrong person, or the wrong year.**
- **Not checking for absurdity.** A negative age, or a child older than the parent, means the setup is wrong.

Speed

- **Draw the three-column table.** Ten seconds, and it removes almost every setup error.
- **Name the younger person as the variable** to avoid fractions.
- **Write the age difference down first.** It is an invariant, and it often solves the problem alone.
- **Back-solve from the choices.** Age problems are ideal for it.
- **Check your answer against every sentence,** not just the one you solved with.

4.8 Percentage word problems

The whole topic is one question: **what is the base?**

PHRASE	THE BASE
x percent of the total	the total
x percent increase	the original value
x percent discount	the marked price
x percent more than B	B
x percent of the remainder	what is left, **not** the original
x percent of the students who passed	the passers, not all students

Multiple stages — chain the multipliers against the right base. In a class of 200 where 60 percent are girls and 25 percent of the girls play a sport: girls are 120, and sporting girls are 30, which is 15 percent of the class, not 25.

Percent of a remainder. "She spends 40 percent on rent, then 25 percent of the remainder on food." The food spend is `0.25 * 0.60 = 15` percent of the original, and she keeps `0.60 * 0.75 = 45` percent.

Population growth is `Initial * (1 + r)^n`, the same structure as compound interest.

Traps

- **Percent of the remainder read as percent of the original.** The biggest trap in the topic.
- **A 20 percent fall needs a 20 percent rise to recover.** It needs 25 percent.
- **Averaging two group percentages.** 25 and 10 percent average to 17.5 only if the groups are equal in size.
- **Percentage points against percent.** A rise from 8 percent to 12 percent is 4 percentage points but a 50 percent increase.
- **Applying a percent to the wrong subgroup.** "30 percent of the women" is not "30 percent of the people".
- **Rounding too early** in a multi-stage chain.

Speed

- **Set the total to 100.** Every answer then reads directly as a percent.
- **Convert everything to multipliers of the original before combining.** "70 percent left, then 80 percent of that" is `0.56` of the original.
- **Draw a two-by-two table** whenever two categories overlap.
- **To reverse a percent change, divide** by the multiplier.
- **Check that the pieces sum to 100 percent** at the end.

4.9 Ratio word problems

The method. Write `ak` and `bk`, then ask: **what stayed the same?** Change only what the story changed, and cross-multiply.

Traps

TRAP	WRONG	RIGHT
Changing both parts when only one moved	10 girls join, giving `(5k+10)/(4k+10)`	`5k/(4k+10)`
Treating ratio numbers as counts	`5 : 4` means 5 boys	`5k` boys
Part-to-part read as part-to-whole	boys are `5/4` of the class	boys are `5/9`
Chaining without matching the shared term	`3:4` and `8:9` give `3:4:9`	scale to `6:8:9`
One multiplier for two independent ratios	incomes `4k:3k`, expenses `3k:2k`	`4x:3x` and `3y:2y`
Direct proportion where it is inverse	16 pumps take longer than 12	pump-hours are fixed, so fewer hours
A count ratio used as a value ratio	coins `3:5:7` gives value ratio `3:5:7`	multiply each count by its denomination

Speed

- **Name k before you read the second sentence.**
- **If a difference is given, use it directly.** `5k - 2k = 600` gives k in one line; do not build the total first.
- **Count the unknowns before solving.** One ratio needs k; two independent ratios need x and y.
- **For inverse proportion, compute the fixed product** — worker-days, pump-hours — and divide.
- **Check against the original ratio as well as the new one.** Both must hold.
- **Sanity-check that counts come out whole.** A fractional k for people means a misread setup.

Section 5 — Geometry

The rule that governs the whole section. GRE geometric figures are **not necessarily drawn to scale**, so appearance is never a safe guide to a length or an angle. You may assume only that lines shown as straight are straight, that points on a line fall in the order shown, and that objects sit in the relative positions shown. A right angle, a pair of parallel lines, an equal pair of sides or a tangent must be **marked or stated** before you may use it.

The exceptions are worth knowing: **coordinate systems, number lines and the graphs in data interpretation are drawn to scale.** Geometric figures are not.

5.1 Lines, angles and transversals

CONCEPT	RULE
Angles on a straight line	sum to 180
Angles around a point	sum to 360
Vertically opposite angles	equal
Complementary angles	sum to **90**
Supplementary angles	sum to **180**

With parallel lines and a transversal

PAIR	RELATIONSHIP	SHAPE TO LOOK FOR
Corresponding	equal	F
Alternate interior	equal	Z
Alternate exterior	equal	
Same-side interior	supplementary	C or U

The shortcut that solves most of these figures. Mark one angle x. Every other angle in the figure is either x or `180 - x`. Decide which from the transversal relationships — corresponding and alternate pairs are equal, same-side interior pairs are supplementary — never from how the angle looks.

Traps

- **Assuming lines are parallel because they look parallel.** They must be marked or stated.
- **Alternate against same-side interior.** Z means equal; C means supplementary.
- **Calling a 90 pair supplementary.** Complementary is 90; supplementary is 180.
- **Measuring off the figure.** A length or an angle that merely looks right is not a given.

Speed

- **Label one angle x and propagate.** The whole figure falls out in seconds.
- **Extend lines** in your sketch to reveal a transversal that was not drawn.
- **Draw an auxiliary parallel line** through a bent vertex between two parallels; it turns a zig-zag into two simple transversal setups.
- **Sum-check at the end:** straight lines to 180, points to 360.

5.2 Triangles

CONCEPT	RULE
Angle sum	180
Exterior angle	equals the **sum of the two remote interior angles**
Area	`(1/2) * base * height`, with the height perpendicular to that base
Triangle inequality	`|b - c| < a < b + c`
Side-angle correspondence	the largest side faces the largest angle
Median	joins a vertex to the midpoint of the opposite side, and halves the area

The triangle inequality in usable form. With sides 5 and 9, the third side x satisfies `4 < x < 14`. The inequalities are strict: `x = 4` or `x = 14` gives a flat, degenerate triangle.

Classification

BY SIDES		BY ANGLES	
Scalene	all different	Acute	all angles below 90
Isosceles	two equal	Right	one angle is 90
Equilateral	all three equal	Obtuse	one angle above 90

Traps

- **Using a slanted side as the height.** The height must be perpendicular to the chosen base.
- **Forgetting the inequality has a lower bound.** Everyone remembers `a + b > c` and forgets `|a - b| < c`.
- **Assuming isosceles or right from the picture.**
- **Assuming the largest side faces the smallest angle.** It is the opposite.
- **Expecting the height to fall inside.** In an obtuse triangle the foot of the altitude can lie beyond the base, and the formula still holds.

Speed

- **Third side: difference `< x <` sum.** One line, always.
- **Exterior angle equals the two remote interiors.** Faster than finding the interior angle first.
- **Area ratios:** same height means areas in the ratio of the bases; same base means areas in the ratio of the heights.
- **Rank the sides to rank the angles, and the reverse.** Many quantitative comparison questions need nothing more.

5.3 The Pythagorean theorem

CONCEPT	RULE
Pythagoras	`leg^2 + leg^2 = hypotenuse^2`
Converse	if `a^2 + b^2 = c^2`, the angle opposite c is 90
Acute test	`a^2 + b^2 > c^2`, with c the longest side
Obtuse test	`a^2 + b^2 < c^2`
Distance in the xy-plane	`sqrt[(x2-x1)^2 + (y2-y1)^2]` — Pythagoras in disguise
Diagonal of a rectangle	`sqrt(l^2 + w^2)`
Space diagonal of a box	`sqrt(l^2 + w^2 + h^2)`

The triples to memorize. Recognizing one saves half a minute every time.

PRIMITIVE	MULTIPLES THAT APPEAR
3, 4, 5	6-8-10, 9-12-15, 12-16-20, 15-20-25, 30-40-50
5, 12, 13	10-24-26, 15-36-39
8, 15, 17	16-30-34
7, 24, 25	
9, 40, 41	
20, 21, 29	

Traps

- **Treating a leg as the hypotenuse.** A negative result under the root means you swapped them.
- **Using the theorem on a non-right triangle.**
- **Assuming a right angle from the picture** with no marked square and no stated fact.
- **Reading `c^2 = a^2 + b^2` as `c = a + b`.** For 3-4-5 that would give 7.
- **Forgetting to take the square root at the end.**

Speed

- **Divide out a common factor and look for a familiar triple.**
- **The hypotenuse is always the biggest number.** Use it to sanity-check.
- **For a right triangle, the area is `(1/2) * leg * leg`** — the legs are perpendicular, so they are base and height.
- **Estimate:** `sqrt(a^2 + b^2)` is always less than `a + b` and more than the larger of a and b.

5.4 Special right triangles

45-45-90 has sides in the ratio `1 : 1 : sqrt(2)`. So the hypotenuse is `leg * sqrt(2)`, and the leg is `hypotenuse / sqrt(2)`. It is isosceles, which is often the missing fact.

30-60-90 has sides in the ratio `1 : sqrt(3) : 2`.

OPPOSITE	SIDE
30	`x`, the shortest
60	`x sqrt(3)`
90	`2x`, the hypotenuse

The anchor: the side opposite 30 is half the hypotenuse. Find it first and everything else is that number times 1, `sqrt(3)` or 2.

Where they come from. A square cut by its diagonal gives two 45-45-90 triangles. An equilateral triangle cut by its altitude gives a 30-60-90.

The derived formulas worth memorizing

FIGURE	FORMULA
Diagonal of a square, side s	`s sqrt(2)`
Area of a square from its diagonal d	`d^2 / 2`
Altitude of an equilateral triangle, side s	`s sqrt(3)/2`
Area of an equilateral triangle, side s	`s^2 sqrt(3)/4`

Traps

- **Putting `sqrt(3)` on the wrong side.** It goes with the 60 angle. In the standard `x : x sqrt(3) : 2x` labeling, the side opposite 30 carries no radical.
- **Multiplying by `sqrt(2)` when you should divide.** Hypotenuse to leg divides.
- **Using `1 : 2 : 3` for a 30-60-90.** The ratio is `1 : sqrt(3) : 2`.
- **Approximating too early.** GRE answers usually contain the radical; keep it symbolic.
- **Confusing the altitude of an equilateral triangle with its side.** The altitude is `s sqrt(3)/2`, always less than s.

5.5 Isosceles and equilateral triangles

CONCEPT	RULE
Base angles	equal, and equal angles imply equal opposite sides
Equilateral angles	60 each, and equiangular implies equilateral for triangles
Base angle from the apex angle A	`(180 - A)/2`
Apex angle from a base angle B	`180 - 2B`
Equilateral altitude	`s sqrt(3)/2`
Equilateral area	`s^2 sqrt(3)/4`
Isosceles right triangle	is the 45-45-90

The symmetry line. In an isosceles triangle, the altitude from the apex to the base is simultaneously the altitude, the median, the perpendicular bisector of the base and the apex angle bisector. **Dropping it is the standard solution move** — it splits the triangle into two congruent right triangles with a known half-base.

Traps

- **Assuming which angle is the apex.** If the given angle is acute, two configurations may be possible, and the answer may be that the relationship cannot be determined.

- **Assuming isosceles from the picture.** Look for tick marks.
- **Using `s^2 sqrt(3)/4` on a merely isosceles triangle.** It is for equilateral only.
- **Taking the base angles to equal the apex.** They equal each other.
- **Equiangular against equilateral.** Equivalent for triangles, but not for other polygons: a rectangle is equiangular and not equilateral.

Speed

- **Drop the altitude immediately.**
- **`(180 - apex)/2` for the base angles.** One step.
- **If two angles are equal, mark the opposite sides equal at once.** It often unlocks the length you need.
- **On "cannot be determined" questions, actively hunt for the second configuration.**

5.6 Similar and congruent figures

Similarity tests, any one is enough: **AA** (two pairs of equal angles), **SSS** (all three side pairs in the same ratio), **SAS** (two side pairs in ratio with the included angles equal). AA is the workhorse, because two equal angle pairs force the third.

Congruence tests: SSS, SAS, ASA, AAS and RHS. **SSA is not a valid test, and neither is AAA** — AAA gives similarity, not congruence.

The scaling rule — the single most important line in this topic

```
lengths and perimeters scale as k
areas scale as k^2
volumes scale as k^3
```

Where similar triangles hide. A line parallel to one side of a triangle cuts off a similar triangle. The altitude to the hypotenuse of a right triangle creates two triangles similar to the original and to each other. Two lines crossing between a pair of parallels make an hourglass.

Setting up the proportion. Write the corresponding vertices in the same order: if triangle ABC is similar to triangle DEF, then `AB/DE = BC/EF = AC/DF`.

Traps

- **Using the linear ratio for areas.** Sides `2:3` give areas `4:9`. The number-one trap in the topic.
- **Using the area ratio for sides.** Areas `16:25` give sides `4:5`.
- **Mismatched correspondence.** Pair the right vertices or the ratio is nonsense.
- **Assuming similarity from appearance.**
- **Treating SSA as a congruence test.**
- **Perimeter ratio.** Perimeters scale like lengths, not like areas.

Speed

- **Look for parallel lines** — they generate similar triangles automatically.
- **Look for a shared angle plus one more equal angle.** That is AA in two seconds.
- **Redraw the two triangles separately, in the same orientation.** Overlapping figures cause most correspondence errors.
- **Shadow, ladder and mirror problems are nearly always similarity.**

5.7 Quadrilaterals

Angles in any quadrilateral sum to 360.

FIGURE	AREA	PERIMETER	DIAGONAL
Rectangle, l and w	`lw`	`2(l + w)`	`sqrt(l^2 + w^2)`
Square, s	`s^2`, or `d^2/2`	`4s`	`s sqrt(2)`
Parallelogram	`base * height`	`2(a + b)`	not equal
Rhombus	`base * height`, or `(d1 * d2)/2`	`4s`	perpendicular
Trapezoid	`(1/2)(a + b) * h`, the parallel sides a and b	sum of four sides	
Kite	`(d1 * d2)/2`		perpendicular

Properties side by side

PROPERTY	PARALLELOGRAM	RECTANGLE	RHOMBUS	SQUARE
Opposite sides equal and parallel	yes	yes	yes	yes
All angles 90	no	yes	no	yes
All sides equal	no	no	yes	yes
Diagonals bisect each other	yes	yes	yes	yes
Diagonals equal	no	yes	no	yes
Diagonals perpendicular	no	no	yes	yes
Diagonals bisect the angles	no	no	yes	yes

Consecutive angles of a parallelogram are supplementary. One angle of 110 forces the next to be 70.

The hierarchy. Every square is a rectangle and a rhombus; every rectangle and rhombus is a parallelogram. The reverse is never guaranteed.

The trapezoid midsegment joins the midpoints of the non-parallel sides and equals `(a + b)/2`. An **isosceles trapezoid** has equal non-parallel sides, equal base angles and equal diagonals.

The optimization fact. Among all rectangles with a fixed perimeter, the **square** has the largest area; among all with a fixed area, the square has the smallest perimeter. A frequent quantitative comparison theme.

Traps

- **Assuming a quadrilateral is a rectangle because it looks like one.**
- **Doubling the side to get the diagonal.** A square's diagonal is about `1.41s`, not `2s`.
- **Assuming equal perimeters mean equal areas,** or the reverse. A 1-by-36 and a 6-by-6 rectangle both have area 36, with perimeters 74 and 24.
- **Taking a parallelogram's diagonals to be equal.** They bisect each other; they are equal only in a rectangle.
- **Taking diagonals to be perpendicular.** True in a rhombus and a kite, not in a general parallelogram.
- **Using a non-parallel side in the trapezoid formula,** or dropping the `1/2`.
- **Adding the diagonals instead of multiplying** in the rhombus area.
- **Units.** Carpeting needs area; fencing needs perimeter.

Speed

- **Square area from the diagonal: `d^2/2`.** Skips the side entirely.
- **Rhombus area from the diagonals: half their product.** Faster than base times height.
- **Say the trapezoid formula as "the average of the parallel sides, times the height".** It sticks.
- **A rectangle's diagonal is often a Pythagorean triple.**
- **The diagonals of a rhombus make four identical right triangles** with legs equal to the half-diagonals, so Pythagoras gives the side instantly.
- **For maximum area or minimum perimeter, go square.**

5.8 Polygons

CONCEPT	FORMULA
Sum of interior angles	`(n - 2) * 180`
Each interior angle, regular	`(n - 2) * 180 / n`
Sum of exterior angles	**360, always**
Each exterior angle, regular	`360 / n`
Interior plus exterior at a vertex	180
Number of diagonals	`n(n - 3)/2`
Diagonals from one vertex	`n - 3`

The exterior-angle sum is 360 for every polygon, whatever n is. This is the most useful fact in the topic: questions about regular polygons are usually faster through the exterior angle.

The values worth knowing

POLYGON	N	INTERIOR SUM	EACH INTERIOR	EACH EXTERIOR
Triangle	3	180	60	120
Quadrilateral	4	360	90	90
Pentagon	5	540	108	72
Hexagon	6	720	120	60
Octagon	8	1080	135	45
Decagon	10	1440	144	36

The regular hexagon bonus. A regular hexagon of side s splits into **six equilateral triangles**, so its area is `6 * s^2 sqrt(3)/4 = 3 s^2 sqrt(3)/2`.

Traps

- **Using `n * 180` instead of `(n - 2) * 180`.**
- **Assuming a polygon is regular.** "Hexagon" alone does not mean equal sides and angles.
- **Thinking the exterior sum grows with n.** It is always 360.
- **Counting sides as diagonals.** A diagonal joins non-adjacent vertices.
- **Applying the "each interior angle" formula to an irregular polygon.** The sum still holds; the individual angles vary.

Speed

- **Go through the exterior angle** whenever a regular polygon's angle is given: `exterior = 180 - interior`, then `n = 360/exterior`. Two steps instead of an equation.
- **Memorize the hexagon:** 120 interior, 60 exterior, six equilateral triangles.
- **Decompose a regular polygon into triangles from the center** for area.

5.9 Circles

CONCEPT	FORMULA
Diameter	`d = 2r`
Circumference	`C = 2 pi r = pi d`
Area	`A = pi r^2`
Arc length, central angle t degrees	`(t/360) * 2 pi r`
Sector area, central angle t degrees	`(t/360) * pi r^2`
Perimeter of a sector	`arc + 2r`
Area straight from circumference	`A = C^2/(4 pi)`

The fraction principle covers arcs and sectors alike. A 90 degree sector is `90/360 = 1/4` of the circle: a quarter of the area and a quarter of the circumference.

ANGLE	FRACTION
30	1/12
45	1/8
60	1/6
90	1/4
120	1/3
180	1/2

Scaling. Double the radius and the circumference doubles but the **area quadruples**. Triple it and the area goes up nine times.

Value of pi: about 3.14, or `22/7` by hand. GRE answers usually keep pi symbolic, so leave it in.

Traps

- **Using the diameter as the radius.** If `d = 14` then `r = 7`. Forgotten constantly.
- **Confusing a chord with an arc.** A chord is straight; an arc is curved and always longer than its chord.
- **Thinking doubling the radius doubles the area.**
- **Confusing sector area with arc length.** One is a piece of `r^2`, the other a piece of r.
- **Forgetting that a sector's perimeter includes the two radii.**
- **Approximating pi too early.**

Speed

- **Find r first, always.** Write it down before touching another formula.
- **Turn the central angle into a fraction of 360** and apply it to the whole-circle quantity.
- **`A = C^2/(4 pi)`** goes from circumference to area with no r step.
- **For shaded regions, outer figure minus inner figure.** Never compute the odd shape directly.

5.10 Chords, tangents and inscribed angles

THEOREM	STATEMENT
Tangent and radius	the radius to the point of tangency is **perpendicular** to the tangent
Perpendicular from the center	a perpendicular from the center to a chord **bisects** it
Equal chords	are equidistant from the center
Inscribed angle	equals **half** the central angle on the same arc
Angle in a semicircle	an angle inscribed in a semicircle is a **right angle**
Same arc	inscribed angles on the same arc are equal
Two tangents from an external point	the tangent segments are equal in length

The chord distance relation. For a chord of length c in a circle of radius r, the perpendicular distance d from the center satisfies

```
r^2 = d^2 + (c/2)^2
```

The semicircle right angle is the most tested of these. If a triangle is inscribed in a circle with one side as the diameter, the opposite angle is 90 — and that is often the unstated key to the whole problem.

Traps

- **Assuming a line is tangent because it looks tangent.**
- **Inverting the inscribed-angle relation.** Inscribed is **half** the central; central is double. Know which way you are going.
- **Forgetting the semicircle right angle.**
- **Assuming any line from the center bisects a chord.** Only the perpendicular one does.
- **Confusing arc measure in degrees with arc length in units.**

Speed

- **Draw the radius to the point of tangency.** It hands you a right angle free.
- **Drop a perpendicular from the center to any chord.** It hands you a right triangle with legs d and `c/2`.

- **Look for a diameter in the figure,** then hunt for the inscribed right angle.
- **Pythagorean triples appear constantly** in chord and tangent problems.

5.11 Inscribed and circumscribed figures

The two facts that cover most of these questions

```
Circle inside a square:   the circle's DIAMETER = the square's SIDE
Square inside a circle:   the square's DIAGONAL = the circle's DIAMETER
```

CONFIGURATION	RELATIONSHIP
Circle inscribed in a square	`d = s`; area ratio circle to square is `pi/4`, about 78.5 percent
Square inscribed in a circle	`diagonal = d`, so `side = d/sqrt(2)` and area `= d^2/2`
Right triangle inscribed in a circle	the **hypotenuse is the diameter**, so `R = hypotenuse/2`
Rectangle inscribed in a circle	the diagonal is the diameter
Circle inscribed in an equilateral triangle, side s	`r = s sqrt(3)/6`
Circle around an equilateral triangle, side s	`R = s sqrt(3)/3`, and note `R = 2r`
Circle inscribed in a right triangle, legs a and b, hypotenuse c	`r = (a + b - c)/2`

Traps

- **Mixing up the two headline facts.** A circle inscribed in a square matches the **side**; the diagonal relationship belongs to the square inscribed in a circle.
- **Losing track of which figure is inside.**
- **Using the radius where the diameter belongs.**
- **Assuming contact when the stem does not say so.** "Inscribed" means touching; merely "inside" gives no relationship at all.
- **Assuming an inscribed triangle is equilateral** without justification.

Speed

- **Write the linking equation in words first** — "the circle's diameter equals the square's side" — then substitute.
- **A right triangle inscribed in a circle hands you the diameter for free.**
- **For area ratios, set `r = 1`.** Circle in a square gives `pi/4`; square in a circle gives `2/pi`.
- **Shaded regions are always outer minus inner.**

5.12 Composite figures

FIGURE	AREA	PERIMETER
Square, s	`s^2`	`4s`
Rectangle	`lw`	`2(l + w)`
Triangle	`(1/2)bh`	`a + b + c`
Parallelogram	`bh`	`2(a + b)`
Trapezoid	`(1/2)(a+b)h`	sum of the four sides
Rhombus	`(d1 d2)/2`	`4s`
Circle	`pi r^2`	`2 pi r`
Sector, angle t	`(t/360) pi r^2`	`arc + 2r`
Regular hexagon, s	`3 s^2 sqrt(3)/2`	`6s`

The procedure. Identify the standard shapes. Decide whether to **add** the pieces or **subtract** one from another. Fill in every dimension on the sketch, including the hidden one you have to deduce. Then compute, and state the units.

A composite figure's perimeter is not the sum of the parts' perimeters. Internal edges are not part of the boundary. **Trace the outline with your finger** — what you trace is what you add.

Scaling. Multiply every length by k and the perimeter scales by k while the area scales by `k^2`.

Traps

- **Including internal edges in the perimeter.**
- **Adding areas where you should subtract.** Read whether the shaded region is inside or outside.
- **Mixing units.** Convert before squaring: 1 square meter is 10,000 square centimeters, not 100.
- **Using a slanted side as a height.**
- **Thinking doubling the dimensions doubles the area.** It quadruples it.

Speed

- **For shaded regions, big minus small.**
- **Look for symmetry.** Many composite figures are two or four copies of one simple shape.
- **Keep pi symbolic** until the final answer.
- **Estimate as a check:** the shaded part should be a plausible fraction of the whole.

5.13 Solids

SOLID	VOLUME	SURFACE AREA
Cube, s	`s^3`	`6 s^2`
Rectangular box	`lwh`	`2(lw + lh + wh)`
Cylinder, r and h	`pi r^2 h`	`2 pi r^2 + 2 pi r h`
Cone, r and h	`(1/3) pi r^2 h`	`pi r^2 + pi r l`, with l the slant height
Sphere, r	`(4/3) pi r^3`	`4 pi r^2`
Hemisphere, r	`(2/3) pi r^3`	`3 pi r^2`, curved plus flat
Prism	`base area * height`	`2(base area) + (perimeter * height)`
Pyramid	`(1/3)(base area)(height)`	base plus lateral faces

Diagonals

```
Face diagonal of a cube, side s:   s sqrt(2)
Space diagonal of a cube, side s:  s sqrt(3)
Space diagonal of a box:           sqrt(l^2 + w^2 + h^2)
Cone slant height:                 l = sqrt(r^2 + h^2)
```

The one-third rule. A cone is one third of the cylinder with the same base and height. A pyramid is one third of the prism with the same base and height.

Scaling. Multiply every length by k and the surface area scales by `k^2` while the volume scales by `k^3`. Doubling **every** dimension of a box multiplies its volume by 8.

Traps

- **Using the diameter as the radius** in cylinder and sphere formulas.
- **Forgetting the `1/3`** for cones and pyramids.
- **Using the height instead of the slant height** in a cone's lateral surface area.
- **Confusing total surface area with curved surface area.** A closed cylinder has two circular ends; an open pipe has none.
- **Thinking doubling every dimension doubles the volume.** It multiplies it by 8. Doubling only one dimension, such as the height, is what doubles it.
- **Mixing units.** 1 cubic meter is 1,000,000 cubic centimeters.
- **Writing a sphere's surface area as `2 pi r^2`.** It is `4 pi r^2`.

Speed

- **Volume of any prism or cylinder is base area times height.** Learn it as one formula, not five.
- **Cone or pyramid? Multiply by one third.**
- **Cube facts together:** face diagonal `s sqrt(2)`, space diagonal `s sqrt(3)`, surface `6s^2`, volume `s^3`.

- **For scaling questions, compute nothing** — just apply `k^2` and `k^3`.
- **Unroll a cylinder's curved surface** into a rectangle of width `2 pi r` and height h, and `2 pi r h` becomes obvious.

Section 6 — Coordinate Geometry

Throughout, this is the **xy-plane**, and a point is written `(x, y)` with x first.

6.1 The plane, reflections and translations

QUADRANT	X	Y	EXAMPLE
I, top right	+	+	`(3, 4)`
II, top left	-	+	`(-3, 4)`
III, bottom left	-	-	`(-3, -4)`
IV, bottom right	+	-	`(3, -4)`

The quadrants run **counterclockwise from the top right**. A point on the x-axis has `y = 0`; a point on the y-axis has `x = 0`; the origin `(0, 0)` belongs to **no** quadrant.

Reflections

REFLECT ACROSS	`(x, y)` BECOMES
the x-axis	`(x, -y)`
the y-axis	`(-x, y)`
the origin	`(-x, -y)`
the line `y = x`	`(y, x)`

The mnemonic: across the x-axis, x stays. The axis you reflect across is the one whose coordinate is unchanged.

Translations. Moving right by a adds a to x; moving up by b adds b to y.

Distance from an axis. The distance of `(x, y)` from the x-axis is `|y|`, and from the y-axis is `|x|`. Always non-negative.

Traps

- **Reversing the coordinates.** `(3, 5)` is not `(5, 3)` unless the point lies on `y = x`.
- **Placing the origin in a quadrant.**
- **Reflecting across the wrong axis.**
- **Assuming one negative coordinate means Quadrant III.** Only if **both** are negative.

Speed

- **Sketch it.** Rough axes with the points marked takes ten seconds and prevents sign errors.
- **For rectangles aligned to the axes, subtract coordinates directly** to get the side lengths.
- **For a triangle with axis-parallel legs, use the two legs as base and height.**

6.2 Distance and midpoint

CONCEPT	FORMULA
Distance between `(x1,y1)` and `(x2,y2)`	`sqrt[(x2-x1)^2 + (y2-y1)^2]`
Midpoint	`((x1+x2)/2, (y1+y2)/2)`
Distance from the origin	`sqrt(x^2 + y^2)`
Horizontal distance, same y	`|x2 - x1|`
Vertical distance, same x	`|y2 - y1|`
Missing endpoint from the midpoint	`B = (2Mx - Ax, 2My - Ay)`

Distance uses differences; midpoint uses sums. That one sentence prevents the two commonest errors in the topic.

"Double the midpoint, subtract the known endpoint" is the missing-endpoint recipe.

Watch for triples. If the horizontal and vertical gaps are 3 and 4, or 6 and 8, or 5 and 12, the distance is a whole number. This happens constantly on the GRE.

Traps

- **Subtracting inside the midpoint formula,** or adding inside the distance formula.
- **Forgetting the square root.** `sqrt(64 + 36) = 10`, not 100.
- **Sign errors with negative coordinates.** `9 - (-1) = 10`, not 8.
- **`sqrt(a^2 + b^2) = a + b`.** Never.
- **Mixing up which point is which.** Harmless for distance, fatal for the missing endpoint.

Speed

- **Draw the right triangle:** horizontal leg, vertical leg, hypotenuse. Then Pythagoras.
- **Look for a triple in the two gaps** before reaching for the calculator.
- **Midpoint is just two averages.**
- **Given two endpoints of a diameter, the center is the midpoint** and the radius is half the distance.

6.3 Slope and intercepts

CONCEPT	FORMULA OR FACT
Slope between two points	`(y2 - y1)/(x2 - x1)` — rise over run
Slope-intercept form	`y = mx + b`
Horizontal line	slope **0**, equation `y = c`
Vertical line	slope **undefined**, equation `x = c`
y-intercept	the y value where `x = 0`
x-intercept	the x value where `y = 0`
For `Ax + By = C`	slope is `-A/B`, y-intercept is `C/B`

Reading the sign. Positive rises left to right; negative falls; zero is horizontal; undefined is vertical.

Steepness is the absolute value. A slope of -4 is steeper than a slope of 2.

Slope is constant along a line, which is exactly how you test three points for collinearity.

Traps

- **Inverting the formula.** The change in **y** goes on top.
- **Mismatched order.** Use the same order for y as for x.
- **Giving a vertical line slope 0.** Horizontal is 0; vertical is undefined.
- **Confusing a negative slope with a small one.** -5 is steep; 0.1 is nearly flat.
- **Reading the intercept off an unsimplified equation.** For `2y = 6x + 10` the intercept is 5.
- **Trusting visual steepness** when the two axes have different scales.

Speed

- **Rewrite everything as `y = mx + b`.** Slope and intercept then read straight off.
- **`slope = -A/B` for `Ax + By = C`.** Instant, with no rearranging.
- **For intercepts, cover the other variable:** set `y = 0` for the x-intercept, `x = 0` for the y-intercept.
- **Equal slopes plus a shared point means collinear; equal slopes with no shared point means parallel.**

6.4 The equation of a line

FORM	EQUATION	USE WHEN
Slope-intercept	`y = mx + b`	you know the slope and the y-intercept
Point-slope	`y - y1 = m(x - x1)`	you know the slope and **any** point
Standard	`Ax + By = C`	the answer choices are in this form
Horizontal	`y = c`	
Vertical	`x = c`	

The two-point workflow. Compute the slope, substitute it and either point into point-slope form, then rearrange into whatever form the choices use.

To test whether a point is on a line, substitute its coordinates and see whether both sides balance. **To find where two lines meet,** solve the two equations together.

Traps

- **Sign error in point-slope.** With the point `(2, -5)`, the left side is `y + 5`.
- **Using the given point as the intercept.** The y-intercept is only the point where `x = 0`.
- **Forgetting to distribute** when expanding `m(x - x1)`.
- **Answer-form mismatch.** Convert before comparing.
- **Assuming the line passes through the origin.** Only if `b = 0`.
- **Vertical lines.** They cannot be written as `y = mx + b` at all.

Speed

- **Use point-slope as your default.** It works from any point and never needs b separately.
- **To check a choice, substitute a known point** rather than deriving the equation.
- **Two points with the same y give a horizontal line; the same x gives a vertical line.**

6.5 Parallel and perpendicular lines

RELATIONSHIP	SLOPE CONDITION
Parallel	`m1 = m2`, with different y-intercepts
Perpendicular	`m2 = -1/m1` — **flip and negate**
The same line	same slope and same intercept
Horizontal and vertical	perpendicular, as a special case

SLOPE	PERPENDICULAR SLOPE
2	`-1/2`
`-3/4`	`4/3`
1	-1
`5/2`	`-2/5`
0, horizontal	undefined, vertical

For a horizontal and a vertical line the product rule does not apply — one slope is undefined — but they are still perpendicular.

Traps

- **The reciprocal without the negative.** The perpendicular to slope 2 is `-1/2`, not `1/2`.
- **The negative without the reciprocal.** It is not -2 either.
- **Assuming equal slopes means the same line.** Different intercepts means parallel and distinct.
- **Forgetting to convert before comparing.** `Ax + By = C` has slope `-A/B`.
- **Perpendicular bisector questions need both the perpendicular slope and the midpoint.** Students routinely supply only one.

Speed

- **Say "flip and negate" every time.** Faster than the product rule.
- **Perpendicularity by inspection:** swap the coefficients of x and y and change one sign. So `2x + 3y = k` is perpendicular to `3x - 2y = k`.
- **Perpendicular bisector recipe:** midpoint, negative reciprocal slope, point-slope form.
- **Sketch it.** If your perpendicular looks parallel, you dropped the negative.

6.6 Parabolas

CONCEPT	FORMULA
Standard form	`y = ax^2 + bx + c`
Vertex form	`y = a(x - h)^2 + k`, vertex at `(h, k)`
Axis of symmetry and vertex x	`x = -b/(2a)`
Vertex y	substitute that x back in
y-intercept	`c`
x-intercepts	solve `ax^2 + bx + c = 0`
Number of x-intercepts	from `D = b^2 - 4ac`

Direction. `a > 0` opens **up** and has a **minimum**. `a < 0` opens **down** and has a **maximum**.

D	X-INTERCEPTS	THE PICTURE
`D > 0`	two	crosses the axis twice
`D = 0`	one	touches the axis at the vertex
`D < 0`	none	floats entirely above or below the axis

Symmetry shortcut. If the roots are `r1` and `r2`, the vertex sits at `x = (r1 + r2)/2`, the midpoint of the roots. Faster than the formula whenever you know the roots.

Transformations of `y = x^2`

CHANGE	EFFECT
`y = x^2 + k`	shifts up by k
`y = (x - h)^2`	shifts **right** by h
`y = -x^2`	flips upside down
`y = ax^2` with `a > 1`	narrower
`y = ax^2` with `0 < a < 1`	wider

Traps

- **Shift direction.** `(x - 3)^2` moves the graph **right** by 3, not left.
- **Sign in the vertex formula.** It is `-b/(2a)`; with `a = 1` and `b = -8` that gives `+4`.
- **Forgetting to substitute back** for the vertex's y-coordinate.
- **Assuming every parabola crosses the x-axis.**
- **Confusing the y-intercept with the vertex.** They coincide only when `b = 0`.
- **Reading "maximum" for an upward parabola.**

Speed

- **Check the sign of a first.** It settles maximum against minimum and often eliminates half the choices.
- **`x = -b/(2a)` is both the axis of symmetry and the vertex x.** One formula, two uses.
- **The y-intercept is just c.** No work at all.
- **For "how many real solutions", compute the discriminant only.**
- **Match graphs to equations by testing `x = 0`** and checking which way it opens.

Section 7 — Statistics and Probability

7.1 Mean, median, mode and range

MEASURE	DEFINITION
Mean	`sum / count`
Sum	`mean * count` — the most useful rearrangement on the test
Median, odd count	the middle value **after sorting**
Median, even count	the average of the two middle values
Mode	the most frequent value
Range	`largest - smallest`

Sort before you take the median. Always. This is the most common careless error in the topic.

What transformations do

TRANSFORMATION	MEAN, MEDIAN, MODE	RANGE AND STANDARD DEVIATION
Add k to every value	each increases by k	**unchanged**
Multiply every value by k	each multiplied by k	multiplied by `|k|`

Mean against median, and skew

- Symmetric data: mean equals median.
- Skewed right, a few very large values: mean **above** median.
- Skewed left, a few very small values: mean **below** median.

Evenly spaced sets: `mean = median = (first + last)/2`. This shortcut appears constantly.

Traps

- **Finding the median without sorting.**
- **Assuming the median is a member of the set.** With an even count it usually is not.
- **Assuming a unique mode.** A set can be bimodal, or have no mode at all.
- **Assuming adding a constant changes the range.** It does not.
- **Averaging the averages.** That works only when the groups are the same size; otherwise you need a weighted average.

Speed

- **`Sum = mean * count`.** Almost every average word problem is solved by turning averages into sums.
- **To find a missing value, work with sums,** not averages.
- **Deviation shortcut:** to average numbers near 200, average their differences from 200 and add 200 back.
- **On "which measure changes" questions, test a tiny concrete set** of three or five numbers rather than reasoning abstractly.
- **The median only cares about position,** so changing an extreme value usually leaves it alone.

7.2 Weighted average

CONCEPT	FORMULA
Two groups	`(n1 a1 + n2 a2)/(n1 + n2)`
Weights as fractions or percents	`w1 a1 + w2 a2`, with `w1 + w2 = 1`
Group sum	`count * average`
Alligation, for the ratio of counts	`n1 : n2 = (a2 - m) : (m - a1)`

Alligation. The ratio of the counts is the **inverse** of the distances from the combined mean. Averages of 60 and 90 combining to 70: the distances are `90 - 70 = 20` and `70 - 60 = 10`, so the counts are in the ratio `20 : 10 = 2 : 1`. Twice as many in the 60 group, which is exactly why the mean sits nearer 60.

The sanity check. A weighted average always lies **between** the two group averages, and **closer to the group with more members.** It can equal them only when the two are equal to begin with. If your answer sits outside that range, you have made an error.

Traps

- **Averaging the averages.** It is the planted wrong answer.
- **Weighting by the wrong quantity.** Weight by the number of items, not by the values.
- **Inverting the alligation ratio.** The larger group is the one whose average is *closer* to the combined mean.
- **Using percentages as weights without normalizing.** They must sum to 1.
- **Omitting a group** when there are three or more.

Speed

- **Convert averages to totals immediately.** These are sum problems in disguise.
- **Use alligation when you need a ratio** rather than a value. A ten-second method.
- **Check the pull direction first.** If the larger group has the higher average, the answer must exceed the plain average — and that alone often identifies the choice.
- **Set the total count to 100** when the problem gives percentages.

7.3 Range and standard deviation

CONCEPT	RULE
Range	`max - min`
Standard deviation	roughly, the average distance from the mean
It is never negative	and it is 0 **only** when every value is identical
Add k to every value	range and standard deviation **unchanged**
Multiply every value by k	both multiplied by `|k|`
Values spread wide	larger standard deviation
Values clustered near the mean	smaller standard deviation

Compare, do not compute. Ask which set has values further from **its own** mean.

```
{10, 10, 10, 10}  ->  standard deviation 0
{9, 10, 10, 11}   ->  small
{2, 6, 14, 18}    ->  large
```

The formula, for understanding rather than for use: `sqrt[ (sum of (each value - mean)^2) / n ]`. The GRE will not ask you to evaluate it under time pressure, but it explains why adding a constant changes nothing — every deviation is unchanged — and why multiplying by k scales it.

Traps

- **Assuming adding a constant increases the spread.** It shifts everything equally.
- **Assuming a bigger mean means a bigger standard deviation.** They are independent.
- **Assuming a bigger range means a bigger standard deviation.** Usually true but **not guaranteed**: the range depends on two values, the standard deviation on all of them.
- **Computing when a comparison would do.**
- **Expecting a negative value.** Impossible.
- **Multiplying by a negative.** It scales by `|k|`, so it stays positive.

Speed

- **"Adding shifts, multiplying scales."** Two words cover every transformation question.
- **Standard deviation 0 means all values are equal.** A frequent quantitative comparison shortcut.
- **The set with more values bunched near the middle has the smaller spread,** even when the extremes match.
- **Range is a two-number question.** Find the maximum and minimum and stop.

7.4 Quartiles, interquartile range and percentiles

TERM	MEANING
Q1	25th percentile, the median of the lower half
Q2	50th percentile, the median
Q3	75th percentile, the median of the upper half
Interquartile range	`Q3 - Q1`
Range	`max - min`

Each quartile section holds 25 percent of the data points. Between Q1 and Q3 lies the middle 50 percent.

Computing them. Sort. Find the median. Then Q1 is the median of the values **below** it and Q3 the median of the values **above** it. For 2, 4, 5, 7, 8, 10, 12, 15: the median is 7.5, Q1 is 4.5, Q3 is 11, and the interquartile range is 6.5.

A boxplot runs from Q1 to Q3 with a line at the median, and the whiskers reach the minimum and maximum.

Percentile is a rank, not a score. Scoring in the 80th percentile means scoring **higher than 80 percent of test takers** — not getting 80 percent of the questions right. Worth knowing for your own score report as much as for the test.

The outlier rule, occasionally tested: a value lying more than `1.5 *` the interquartile range below Q1 or above Q3.

Traps

- **Percentile read as a percentage score.**
- **Assuming quartiles are evenly spaced numerically.** Each holds 25 percent of the *data points*, not 25 percent of the numeric range.
- **Forgetting to sort.**
- **Including the median in both halves.** With an odd count, exclude it when finding Q1 and Q3.
- **Computing the interquartile range as `Q3 + Q1`.**
- **Assuming the 50th percentile is the mean.** It is the median.

Speed

- **Sort first, then count positions.** These are counting exercises, not arithmetic.
- **Q1 and Q3 are just medians of the halves** — the same procedure twice.
- **Percentile to count: multiply the percentile by the total.**
- **A long whisker with a short box means skew.**

7.5 Distributions and the normal curve

SHAPE	RELATIONSHIP	THE PICTURE
Symmetric	mean = median	balanced bell
Skewed **right**	mean **above** median	long tail to the right
Skewed **left**	mean **below** median	long tail to the left

The memory hook: the tail pulls the mean. It settles every skew question in three seconds.

The normal distribution is perfectly symmetric and bell-shaped, with mean, median and mode all at the center, and it is described completely by its mean and standard deviation.

The 68-95-99.7 rule

WITHIN	PERCENT OF THE DATA
1 standard deviation of the mean	about 68
2 standard deviations	about 95
3 standard deviations	about 99.7

Because the curve is symmetric, half of each band sits on each side. So about **34** percent lies between the mean and 1 standard deviation above it, about **13.5** percent between 1 and 2, and about **2.35** percent between 2 and 3. Those three numbers cover almost every GRE normal question.

Frequency tables. When data arrives as values with frequencies, the mean is a weighted average: multiply each value by its frequency, sum, and divide by the total frequency.

Traps

- **Reversing the skew direction.** Skewed right means the tail points right and the mean exceeds the median, even though most of the data sits on the left.
- **Applying 68-95-99.7 to data that is not normal.** The stem must say normally distributed.
- **Forgetting to halve.** "5 percent outside 2 standard deviations" means 2.5 percent in each tail.
- **Reading a frequency as a value.** Multiply; do not add frequencies to values.
- **Confusing the curve's height with probability.** It is the **area** that gives the proportion.

Speed

- **Sketch the bell curve** and mark the mean and the standard deviation boundaries. Every normal question then reads straight off the sketch.
- **Symmetry halves the work:** the area below the mean is always 50 percent.
- **For frequency tables, keep a running total** of value times frequency.

7.6 Counting and permutations

CONCEPT	FORMULA
Fundamental counting principle	multiply the choices at each stage
Factorial	`n! = n(n-1)...(1)`, and `0! = 1`
Arrangements of n distinct items	`n!`
Permutations of r from n	`P(n,r) = n!/(n-r)!`
Arrangements with repeats	`n! / (a! b! c! ...)`
Circular arrangements of n	`(n-1)!`

Factorials to know: `1, 2, 6, 24, 120, 720, 5040` for 1! through 7!.

Repeated letters. BANANA has 6 letters with A three times and N twice, so `6!/(3! * 2!) = 60` distinct arrangements.

The two restriction techniques. Glue when items must be together: treat the block as one item, arrange, then multiply by the internal arrangements of the block. **Complement** when items must be apart: total arrangements minus those where they are together.

Circular arrangements lose one degree of freedom, because rotations are the same arrangement.

Traps

- **Using a permutation where order does not matter.** "How many committees of 3" is a combination.
- **Forgetting to divide by the repeats.**
- **Adding instead of multiplying.** "And" multiplies; only mutually exclusive alternatives, "or", add.
- **Treating `0!` as 0.** It is 1.
- **Counting circular arrangements as `n!`.**
- **Double counting** in restriction problems — glue **or** complement, not both at once.
- **Assuming repetition is or is not allowed.** A three-digit code from 0 to 9 allows repeats, giving 1000, unless the stem says the digits are distinct, which gives `10 * 9 * 8 = 720`.

Speed

- **Ask "does order matter?" before writing anything.**
- **Draw slots.** One blank per position, fill in the number of choices, multiply. This handles most GRE counting questions with no formula at all.
- **"At least one" almost always means the complement:** total minus none.
- **Cancel factorials rather than expanding.** `10!/8! = 90`.
- **Check the size.** Counting answers grow fast; a suspiciously small one usually means you added where you should have multiplied.

7.7 Combinations

CONCEPT	FORMULA
Combinations of r from n	`C(n,r) = n!/[r!(n-r)!]`
Symmetry	`C(n,r) = C(n, n-r)`
Choose none or all	`C(n,0) = C(n,n) = 1`
Choose one	`C(n,1) = n`
Choose two	`C(n,2) = n(n-1)/2`
Total subsets of an n-element set	`2^n`

The symmetry rule saves real time. `C(10,8) = C(10,2) = 45`. Always convert to the smaller r.

	VALUE
`C(5,2)`	10
`C(6,2)`	15
`C(8,2)`	28
`C(10,2)`	45

`C(n,2) = n(n-1)/2` covers handshakes, pairings, line segments between points and polygon diagonals. It is the single most reusable counting formula on the test.

Separate pools multiply. Choosing 2 from 5 men and 3 from 6 women gives `C(5,2) * C(6,3) = 10 * 20 = 200`.

Traps

- **Using permutations for selections.** A committee of 3 from 9 is 84, not `P(9,3) = 504`.
- **Forgetting the symmetry shortcut** and grinding `C(12,10)` the long way.
- **Adding when you should multiply** across separate pools.
- **Summing "at least" cases and missing one.** Use the complement.
- **Treating identical items as distinct, or the reverse.**
- **Choosing three in order for a ranked list.** That **is** a permutation — re-read whether the roles differ.

Speed

- **Order matters, use P. Order does not, use C.** Decide before writing.
- **Use symmetry to make r as small as possible.**
- **Cancel before multiplying.** Never compute `9!` outright.
- **"At least one" means total minus none.**
- **Sanity check:** `C(n,r)` is always a whole number and never exceeds `P(n,r)`.

7.8 Basic probability

CONCEPT	RULE
Range	`0 <= P <= 1`
Certain	`P = 1`
Impossible	`P = 0`
Complement	`P(not A) = 1 - P(A)`
All outcomes	sum to 1
Odds in favor of `a : b`	`P = a/(a + b)`

The complement rule is the workhorse. Anything containing "at least one" should trigger it: `P(at least one) = 1 - P(none)`.

The standard totals

EXPERIMENT	TOTAL OUTCOMES
One coin	2
Two coins	4
n coins	`2^n`
One die	6
Two dice	**36**
A standard deck	52 cards, 4 suits of 13, 26 red, 26 black, 12 face cards

Two-dice sums. The most likely sum is 7, with 6 ways out of 36. The sums 2 and 12 have one way each. For a sum s from 2 to 7 the count is `s - 1`, and it mirrors downward after 7.

Traps

- **The wrong denominator.** Two dice give 36 outcomes, not 11 or 12.
- **Forgetting that `(3,6)` and `(6,3)` are different outcomes.**
- **A probability above 1.** A sure sign of a counting error.
- **Adding probabilities of events that can both happen.** Subtract the overlap.
- **Assuming replacement.** "Two marbles are drawn" conventionally means without replacement.
- **Confusing odds with probability.** Odds of `3 : 2` in favor means a probability of `3/5`.

Speed

- **Count the total first.** Get the denominator right and the rest usually follows.
- **"At least one" means 1 minus none.** Automatic.
- **For sequential draws, multiply stage by stage,** adjusting the counts each time.
- **Simplify as you go.** Answer choices are in lowest terms.
- **Sanity-check the size:** a likely event should come out well above `1/2`.

7.9 Compound probability, independence and dependence

SITUATION	RULE
A **and** B, independent	`P(A) * P(B)`
A **and** B, dependent	`P(A) * P(B given A)`
A **or** B, mutually exclusive	`P(A) + P(B)`
A **or** B, in general	`P(A) + P(B) - P(A and B)`
Not A	`1 - P(A)`
At least one	`1 - P(none)`
Neither A nor B	`1 - P(A or B)`
Test for independence	`P(B given A) = P(B)`
With replacement	independent
Without replacement	dependent

Independent and mutually exclusive are different ideas, and students conflate them constantly. Independence is about **multiplication**: one event does not change the other's probability. Mutual exclusivity is about **addition**: the two cannot both happen. In fact two events with non-zero probability cannot be both — mutually exclusive events are strongly **dependent**, because if one happens the other definitely does not.

Conditional probability. `P(B given A)` is the probability of B once A has happened, so `P(A and B) = P(A) * P(B given A)`, and rearranged, `P(B given A) = P(A and B) / P(A)`.

With or without replacement. With replacement the pool resets and the draws are independent. Without replacement, **both** the favorable count and the total shrink.

The gambler's fallacy. Independent trials have no memory. After five heads, the probability of a head on the sixth toss is still `1/2`. The GRE tests this directly.

Traps

- **Adding when you should multiply.** "And" multiplies.
- **Forgetting to subtract the overlap** in an "or" question. Double counting is the classic error, and a total above 1 is the tell.
- **Treating dependent draws as independent.**
- **Assuming mutual exclusivity** before checking for overlap.
- **Order in "one red and one blue" problems.** If no order is specified, both count: `(4/10)(6/9) + (6/10)(4/9)`.
- **Confusing `P(A and B)` with `P(B given A)`.** The conditional already assumes A happened, so its denominator is smaller.

Speed

- **Underline the connective word:** and, or, at least, neither.
- **Ask: does the pool change?** One question settles independent against dependent.
- **Draw the tree** for two-stage problems. Each path multiplies; add the paths that qualify.
- **Write the fractions before simplifying** so the shrinking denominators stay visible.
- **For conditional probability, restrict the universe.** "Given that the student studies Spanish" makes the Spanish students your new total.
- **Sanity check:** without replacement, the probability of repeating the same color is always **lower** than with replacement.

7.10 Overlapping sets

Two groups

```
In A or B        = A + B - (in both)
Total group      = (A only) + (B only) + (both) + (neither)
A only           = A - (both)
Neither          = Total - (A or B)
At least one     = Total - neither
```

Three groups

```
In at least one = A + B + C - (A and B) - (B and C) - (A and C) + (A and B and C)
```

The final term is **added back** because members of all three were subtracted one time too many.

The two-by-two table, for two yes-or-no attributes

	B YES	B NO	TOTAL
A yes	both	A only	A total
A no	B only	neither	not-A total
Total	B total	not-B total	grand total

Every row and column must add up, so **filling in four of the nine numbers determines the rest** — usually the grand total, the two category totals and the overlap. That self-checking property is why the table beats a Venn diagram for two-group problems.

Traps

- **Forgetting to subtract the overlap.** 25 plus 18 giving 43 in a class of 40 is impossible, and that impossibility should be your signal.
- **"A only" read as "A".** The 25 cricket players include the 8 who also play football, so cricket-only is 17.
- **Forgetting the "neither" category.** It belongs to the total but not to "A or B".
- **Sign error in the three-group formula.** The pairs subtract; the triple adds back.
- **Assuming the categories are exhaustive.** They rarely are.

Speed

- **Use the two-by-two table for two groups.** It builds an arithmetic check into every row and column.
- **Fill in the overlap first,** then work outward by subtraction.
- **Set the total to 100** for percentage problems.
- **For three groups, fill the Venn diagram from the center outward:** all three, then the pairs minus the center, then the singles.

- **Sanity check:** no region can be negative, and the parts must sum to the total.

Section 8 — Data Interpretation

Every Quant section contains a data set. These questions are not hard mathematically; they are won and lost on reading the display correctly and on not computing more than you need.

8.1 The 40-second orientation

Before reading a single question, check five things. Forty seconds spent here saves two minutes across the set.

1. **Title** — what is being measured?
2. **Axes and column headers** — what are the variables?
3. **Units** — thousands, millions, percent? **The single most common trap.**
4. **Scale** — does the axis start at zero? A truncated axis exaggerates differences.
5. **Footnotes** — the small print often redefines something, and it usually matters for at least one question.

The recurring calculations

QUESTION TYPE	METHOD
Percent of total	`part/total * 100`
Percent change	`(new - old)/old * 100` — **always divide by the earlier value**
Ratio of two values	divide
Average across categories	`sum / count`
"How many times greater"	divide
"Approximately"	estimate, after checking the answer spread

Traps

- **Ignoring the units.** A y-axis reading "revenue in thousands" means a bar at 45 is 45,000.
- **Trusting a truncated axis.** A chart starting at 90 makes a 5 percent difference look enormous. Read the values, not the picture.
- **Percent of the wrong total.** "What percent of Region A's sales" is not "what percent of total sales".
- **Confusing percent change with percentage-point change.**
- **Reading the adjacent bar, row or year.** The classic time-pressure error.
- **Extrapolating.** Unless the question says to project, do not assume a trend continues.
- **Over-precision.** If the stem says "approximately", estimate.

Speed

- **Read the display before the questions.**
- **Check the answer spread before computing.** Widely spaced choices mean estimation is intended.
- **Write the extracted numbers down** rather than holding them in your head while calculating.
- **For "which is greatest", compare visually to shortlist,** then verify only the top two.
- **Compute a shared total once** and reuse it across all the questions in the set.
- **The calculator is there, but reaching for it costs time.** `180/600` is a mental step: rewrite awkward numbers as fractions, so `180/600 = 3/10 = 30` percent.

8.2 Bar graphs and line graphs

BAR TYPE	WHAT IT SHOWS
Simple	one value per category
Grouped	several series side by side per category
Stacked	components summing to a total per category

The topic's most reliable trap. In a **stacked** bar, a middle segment's value is the **difference between its top and bottom edges**, not the height of its top edge.

Reading a line graph

FEATURE	MEANING
Steep upward slope	rapid increase
Flat segment	no change
Steepest segment	greatest rate of change
Highest point	maximum **value**, not maximum growth
Two lines crossing	the two quantities are equal there
Vertical gap between lines	their difference at that moment

Maximum value against maximum increase. A line can reach its highest point in a year when it grew only slightly. "Largest increase" means the steepest segment; "largest value" means the highest point. GRE questions deliberately test both.

Traps

- **Reading a stacked segment's top edge as its value.**
- **A non-zero axis origin.**
- **Confusing greatest value with greatest increase.**
- **Misreading which series is which** in a grouped chart. Check the legend before every question.
- **Assuming equal intervals on the horizontal axis.** Occasionally the years skip.
- **Interpolating between points** as if the data were continuous. The line joins measured points; intermediate values are not data.

Speed

- **Mark the legend before you start.** A five-second habit that prevents series mix-ups.
- **For "greatest change", look for the steepest segment** rather than computing every difference.
- **For stacked bars, write each segment's value in the margin once** and reuse it across the set.
- **Estimate to the nearest gridline** when the stem says "approximately".

8.3 Pie charts

CONCEPT	FORMULA
Slice value	`(percent/100) * total`
Percent from a slice	`slice/total * 100`
Degrees from percent	`percent * 3.6`
Percent from degrees	`degrees / 3.6`
All slices together	100 percent, or 360 degrees

One percent is 3.6 degrees. Memorize it and degree questions take five seconds.

PERCENT	DEGREES
10	36
20	72
25	90
33 1/3	120
50	180

The headline trap: percentages from two pies with different totals are not comparable. Convert both to absolute values first. Company A with a total of 50 million spending 30 percent on research spends 15 million; Company B with a total of 120 million spending 20 percent spends 24 million. B spends more despite the lower percentage.

A missing slice is 100 percent minus the rest.

Traps

- **Comparing percentages across pies with different totals.**
- **Treating a slice as an absolute number.** It is a share until you multiply by the total.

- **"Percent of a slice".** "What percent of the staff budget goes to training" uses the staff amount as the base, not the whole budget.
- **Confusing degrees with percent.** 25 percent is 90 degrees.
- **Assuming the visual slice size is exact.** Read the labels.

Speed

- **Convert every percent to a value once**, at the start of the set, and reuse.
- **For "which is larger" across two pies, always multiply out.**
- **Halves and quarters by eye.** A slice visibly bigger than a quarter exceeds 25 percent, which is often enough to eliminate choices.
- **Watch for "of the X category" wording** — it changes the base.

8.4 Tables

Spend 30 seconds on the row labels, the column headers, the units in every **column separately, whether a totals row is provided or must be computed, and the footnotes.**

QUESTION	METHOD
Row or column total	sum across or down
Percent of total	`cell / the relevant total`
Percent change across columns	`(later - earlier)/earlier`
Ratio between two cells	divide
Average of a row	`row total / number of entries`
Which is greatest	scan, shortlist, then compute

Cross-referencing is where tables bite. Many questions need two cells from different rows and columns. **Locate both and write them down before computing** — that single habit prevents the wrong-row error.

Traps

- **Reading the adjacent row or column.**
- **Mixed units across columns.** One in thousands, another in units.
- **Using the grand total where a row total is needed,** or the reverse.
- **Assuming the listed rows sum to the total row.** Some tables carry an unlisted "other" category, and a footnote will say so.
- **Ignoring a footnote.**
- **Computing more than needed.** If the question concerns two rows, do not total the table.

Speed

- **Locate, then calculate.**
- **Compute a shared total once** and reuse it.
- **Estimate to shortlist** on "which is greatest", then compute only the finalists.
- **For per-unit comparisons such as average salary or cost per item, divide the totals** rather than working entry by entry.

8.5 Multi-display sets

COMBINATION	WHAT EACH DISPLAY SUPPLIES
Pie plus bar	the pie gives shares, the bar gives the totals to multiply by
Table plus line graph	the table gives absolute values, the line gives the trend
Two bar charts	one gives volume, the other gives price or rate
Stacked bar plus pie	the stacked bar gives composition over time, the pie gives one year in detail

The three-step method. Identify what the question wants — a value, a percent, a ratio or a change. Decide which display holds each ingredient and write them down separately. Then combine, usually by multiplying.

The key conversion

```
absolute value = share (from the pie) * total (from the bar or table)
```

Percent change across two displays. Build both years' absolute values first, then apply `(new - old)/old`. Do **not** try to combine the share's percent change with the total's in your head.

Multipliers compose, percent changes do not add. A 25 percent rise in the share alongside a 20 percent rise in the total gives `1.25 * 1.20 = 1.50`, a **50** percent rise, not 45. If the share stays constant, the category's percent change simply equals the total's.

Traps

- **Applying one year's pie chart to a different year.** Shares change unless stated otherwise.
- **Adding percent changes.**
- **Using the wrong total** when two bar charts are present.
- **Mixing units across displays.**
- **Answering from one display when both are needed.** That is the whole design of these sets — if a question feels unanswerable, you are probably missing the second display.
- **Assuming a relationship the data does not state,** such as constant shares or linear growth.

Speed

- **Label each display in the margin once** with what it supplies: "shares", "totals".
- **Convert shares to absolute values once,** then answer the whole set from your converted numbers.
- **Write intermediate values down.** Holding four numbers from two charts in your head is exactly where the errors happen.

Section 9 — The 25 traps that cost the most marks

Every one of these is an error students make **after** they have understood the topic. If you read nothing else the night before, read this page.

Reading the question

1. **Solving for x when the question asked for `2x`, or for `x + y`.** Underline what is asked before you start, and read the last line again before you answer.
2. **Answering a multiple-answer question with one selection.** Square boxes mean select all. Check the button shape before you answer.
3. **Stopping after two correct options.** There may be a third, and there is no partial credit.
4. **Rounding when no rounding was asked for, or not rounding when it was.** Numeric entry has no safety net.

Assuming what was not stated

5. **Assuming a variable is a positive integer.** Unless the stem says so, x can be negative, zero or a fraction. "Number" does not mean "integer".
6. **Assuming a figure is to scale in quantitative comparison,** or assuming a right angle, parallel lines or equal sides that were never marked.
7. **Choosing "cannot be determined" when both quantities are pure numbers.** Impossible: the relationship is fixed even if you cannot see it.
8. **Testing only one value on quantitative comparison.** One test can rule things out; it can never prove a relationship. Run the FROZEN check.

Algebra

9. **`(a + b)^2 = a^2 + b^2`.** The `2ab` is missing. The most common algebra error on the test.
10. **Losing the negative root.** `x^2 = 49` gives 7 **and** -7. And `sqrt(x^2) = |x|`, not x.
11. **Dividing by a variable that could be zero.** From `x^2 = 3x`, dividing by x loses `x = 0`. Factor instead.
12. **Forgetting to flip the inequality** when multiplying or dividing by a negative — and dividing by a variable whose sign you do not know.
13. **Sum of the roots written as `b/a`.** It is `-b/a`.
14. **Canceling across a plus sign.** `(x + 3)/3` is not x. You may cancel factors, never terms.

Percent, ratio and rate

15. **Percent change computed on the new value.** Always divide by the **original**.
16. **Adding successive percents.** Multiply the multipliers: a 20 percent fall then a 20 percent rise leaves you 4 percent down, and recovering a 20 percent fall needs a 25 percent rise.
17. **"Percent of the remainder" read as "percent of the original".**
18. **Part-to-part read as part-to-whole.** `2 : 3` means `2/5` of the whole, not `2/3`.
19. **Averaging the speeds.** Average speed is total distance over total time: add the distances and add the times, never the speeds. In work problems it is the other way round — there the **rates** add and the times do not.
20. **Averaging two averages of unequal groups.** You need a weighted average, and the result always sits closer to the bigger group.

Geometry, counting and data

21. **Using the linear ratio for areas.** Sides `2 : 3` give areas `4 : 9` and volumes `8 : 27`. Likewise, doubling the radius quadruples the area.
22. **Using the diameter as the radius,** or using a slanted side as a height.
23. **Using a permutation where order does not matter** — and forgetting that "at least one" means one minus the probability of none.
24. **Forgetting to subtract the overlap.** `P(A or B)` subtracts `P(A and B)`, and two groups overlap unless you are told otherwise. A probability above 1 or a group total above the class size is the tell.
25. **Ignoring the units on a data display,** reading a stacked segment's top edge as its value, or comparing percentages across two pie charts with different totals.

Appendix — Memorize this

Everything on this page should be recall, not calculation. If any line makes you pause, that is the line to drill.

Squares, 1 to 30

```
 1    4    9   16   25   36   49   64   81  100
121  144  169  196  225  256  289  324  361  400
441  484  529  576  625  676  729  784  841  900
```

Cubes, 1 to 15

```
  1     8    27    64   125   216   343   512
729  1000  1331  1728  2197  2744  3375
```

Powers worth knowing

BASE	POWERS
2	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096
3	3, 9, 27, 81, 243, 729
4	4, 16, 64, 256, 1024
5	5, 25, 125, 625, 3125

The 25 primes below 100

```
 2   3   5   7  11  13  17  19  23  29
31  37  41  43  47  53  59  61  67  71
73  79  83  89  97
```

Remember: **1 is not prime, and 2 is the only even prime.**

Fraction, decimal and percent

FRACTION	DECIMAL	PERCENT
1/2	0.5	50
1/3	0.333...	33.33
2/3	0.666...	66.67
1/4	0.25	25
3/4	0.75	75
1/5	0.2	20
2/5	0.4	40
3/5	0.6	60
4/5	0.8	80
1/6	0.1666...	16.67
5/6	0.8333...	83.33
1/7	0.142857...	14.29
1/8	0.125	12.5
3/8	0.375	37.5
5/8	0.625	62.5
7/8	0.875	87.5
1/9	0.111...	11.11
1/10	0.1	10
1/11	0.0909...	9.09
1/12	0.0833...	8.33
1/16	0.0625	6.25
1/20	0.05	5
1/25	0.04	4

Why this table earns its place: dividing by 8 is faster than multiplying by 0.125, and spotting that 37.5 percent is `3/8` turns an awkward multiplication into a cancellation.

Roots and constants

VALUE	APPROXIMATION
`sqrt(2)`	1.414
`sqrt(3)`	1.732
`sqrt(5)`	2.236
`sqrt(6)`	2.449
`sqrt(7)`	2.646
`sqrt(8)`	2.828
`sqrt(10)`	3.162
`pi`	3.142, or `22/7`

Pythagorean triples

PRIMITIVE	MULTIPLES THAT SHOW UP
3, 4, 5	6-8-10, 9-12-15, 12-16-20, 15-20-25, 30-40-50
5, 12, 13	10-24-26, 15-36-39
8, 15, 17	16-30-34
7, 24, 25	
9, 40, 41	
20, 21, 29	

Special triangles and the figures built from them

```
45-45-90            sides  1 : 1 : sqrt(2)
30-60-90            sides  1 : sqrt(3) : 2    (the side opposite 30 is half the hypotenuse)

Square, side s      diagonal  s sqrt(2)        area from the diagonal  d^2 / 2
Equilateral, side s altitude  s sqrt(3)/2      area  s^2 sqrt(3)/4
Regular hexagon, s  area  3 s^2 sqrt(3)/2      (six equilateral triangles)
Cube, side s        face diagonal s sqrt(2)    space diagonal s sqrt(3)
```

Angles in polygons

POLYGON	N	INTERIOR SUM	EACH INTERIOR	EACH EXTERIOR
Triangle	3	180	60	120
Quadrilateral	4	360	90	90
Pentagon	5	540	108	72
Hexagon	6	720	120	60
Octagon	8	1080	135	45
Decagon	10	1440	144	36

The exterior angles of any polygon sum to 360, always.

Factorials and small combinations

```
1! = 1     2! = 2     3! = 6     4! = 24
5! = 120   6! = 720   7! = 5040  8! = 40320        and 0! = 1
```

	VALUE
`C(5,2)`	10
`C(6,2)`	15
`C(8,2)`	28
`C(10,2)`	45

`C(n,2) = n(n-1)/2` — handshakes, pairings, segments between points, polygon diagonals.

Probability totals

EXPERIMENT	TOTAL OUTCOMES
n coins	`2^n`
One die	6
Two dice	36, with 7 the most likely sum at 6 ways
A deck of cards	52 cards, 4 suits of 13, 26 red, 12 face cards

The normal curve

```
within 1 standard deviation  about 68 percent
within 2                     about 95 percent
within 3                     about 99.7 percent

between consecutive marks:   34  /  13.5  /  2.35  percent
```

Conversions and small facts

```
1 hour = 3600 seconds                1 kilometer = 1000 meters
kilometers per hour to meters per second:  multiply by 5/18   (back again: 18/5)
1 square meter = 10,000 square centimeters
1 cubic meter  = 1,000,000 cubic centimeters
1 percent of a pie chart = 3.6 degrees
doubling time under compounding, roughly:  72 / (rate as a percent)
```

The two ordering chains

```
If 0 < x < 1:   x^2  <  x  <  sqrt(x)  <  1/x
If x > 1:       1/x  <  sqrt(x)  <  x  <  x^2
```

Units-digit cycles for powers

UNITS DIGIT OF THE BASE	CYCLE	LENGTH
0, 1, 5, 6	itself	1
4	4, 6	2
9	9, 1	2
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4

One last page before the exam

```
Every question is worth the same mark.
Two passes. Easy points first.
30 seconds to find a route, or flag it and move on.
Never leave a blank — there is no penalty.
Read the last line again before you answer.
Write every intermediate value down.
Check the shape of the answer: no negative lengths,
no probability above 1, no average outside the data.
A hard question tells you nothing. Keep going.
```

Good luck. You have done the work — now just do the test in front of you.

WHAT TO DO NEXT

You have the formulas. Now build the speed.

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